



MKS Type DELTA™ II Flow Ratio Controller (Devicenet®)

Preliminary Instruction Manual

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**MKS Type DELTA™ II
Flow Ratio Controller
(Devicenet®)**

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Safety Information

Symbols Used in This Instruction Manual

Definitions of WARNING, CAUTION, and NOTE messages used throughout the manual are:



Warning

The **WARNING** sign denotes a hazard. It calls attention to a procedure, practice, condition, or the like, which, if not correctly performed or adhered to, could result in injury to personnel.



Caution

The **CAUTION** sign denotes a hazard. It calls attention to an operating procedure, practice, or the like, which, if not correctly performed or adhered to, could result in damage to or destruction of all or part of the product.



Note

The **NOTE** sign denotes important information. It calls attention to a procedure, practice, condition, or the like, which is essential to highlight.

Symbols Found on the Unit

The following table describes symbols that may be found on the unit.

Table 1: Definition of Symbols Found on the Unit

 On (Supply) IEC 417, No. 5007	 Off (Supply) IEC 417, No. 5008	 Earth (ground) IEC 417, No. 5017	 Protective Earth (ground) IEC 417, No. 5019
 Frame or Chassis IEC 417, No. 5020	 Equipotentiality IEC 417, No. 5021	 Direct Current IEC 417, No. 5031	 Alternating Current IEC 417, No. 5032
 Both Direct and Alternating Current IEC 417, No. 5033-a	 Class II Equipment IEC 417, No. 5172-a	 Three Phase Alternating Current IEC 617-2, No. 020206	
 Caution (refer to accompanying documents) ISO 3864, No. B.3.1	 Caution, Risk of Electric Shock ISO 3864, No. B.3.6	 Caution, Hot Surface IEC 417, No. 5041	

Safety Procedures and Precautions

Observe the following general safety precautions during all phases of operation of this instrument. Failure to comply with these precautions or with specific warnings elsewhere in this manual violates safety standards of intended use of the instrument and may impair the protection provided by the equipment. MKS Instruments, Inc. assumes no liability for the customer's failure to comply with these requirements.

DO NOT SUBSTITUTE PARTS OR MODIFY INSTRUMENT

Do not install substitute parts or perform any unauthorized modification to the instrument. Return the instrument to an MKS Calibration and Service Center for service and repair to ensure that all safety features are maintained.

SERVICE BY QUALIFIED PERSONNEL ONLY

Operating personnel must not attempt component replacement and internal adjustments. Any service must be made by qualified service personnel only.

USE CAUTION WHEN OPERATING WITH HAZARDOUS MATERIALS

If hazardous materials are used, observe the proper safety precautions, completely purge the instrument when necessary, and ensure that the material used is compatible with the wetted materials in this product, including any sealing materials.

PURGE THE INSTRUMENT

After installing the unit, or before removing it from a system, purge the unit completely with a clean, dry gas to eliminate all traces of the previously used flow material.

USE PROPER PROCEDURES WHEN PURGING

This instrument must be purged under a ventilation hood, and gloves must be worn for protection.

DO NOT OPERATE IN AN EXPLOSIVE ENVIRONMENT

To avoid explosion, do not operate this product in an explosive environment unless it has been specifically certified for such operation.

USE PROPER FITTINGS AND TIGHTENING PROCEDURES

All instrument fittings must be consistent with instrument specifications, and compatible with the intended use of the instrument. Assemble and tighten fittings according to manufacturer's directions.

CHECK FOR LEAK-TIGHT FITTINGS

Carefully check all vacuum component connections to ensure leak-tight installation.

OPERATE AT SAFE INLET PRESSURES

Never operate at pressures higher than the rated maximum pressure (refer to the product specifications for the maximum allowable pressure).

INSTALL A SUITABLE BURST DISC

When operating from a pressurized gas source, install a suitable burst disc in the vacuum system to prevent system explosion should the system pressure rise.

KEEP THE UNIT FREE OF CONTAMINANTS

Do not allow contaminants to enter the unit before or during use. Contamination such as dust, dirt, lint, glass chips, and metal chips may permanently damage the unit or contaminate the process.

ALLOW THE UNIT TO WARM UP

If the unit is used to control dangerous gases, they should not be applied before the unit has completely warmed up. Use a positive shutoff valve to ensure that no erroneous flow can occur during warm up.

Sicherheitshinweise für das Massenflussgerät

In dieser Betriebsanleitung vorkommende Symbole

Bedeutung der mit WARNUNG!, VORSICHT! und HINWEIS gekennzeichneten Absätze in dieser Betriebsanleitung.



Warnung!

Das Symbol **WARNUNG!** weist auf eine Gefahr für das Bedienpersonal hin. Es macht auf einen Arbeitsablauf, eine Arbeitsweise, einen Zustand oder eine sonstige Gegebenheit aufmerksam, deren unsachgemäße Ausführung bzw. ungenügende Berücksichtigung zu Verletzungen führen kann.



Vorsicht!

Das Symbol **VORSICHT!** weist auf eine Gefahr für das Gerät hin. Es macht auf einen Bedienungsablauf, eine Arbeitsweise oder eine sonstige Gegebenheit aufmerksam, deren unsachgemäße Ausführung bzw. ungenügende Berücksichtigung zu einer Beschädigung oder Zerstörung des Gerätes oder von Teilen des Gerätes führen kann.



Hinweis

Das Symbol **HINWEIS** macht auf wichtige Informationen bezüglich eines Arbeitsablaufs, einer Arbeitsweise, eines Zustands oder einer sonstige Gegebenheit aufmerksam.

Erklärung der am Gerät angebrachten Symbole

Nachstehender Tabelle sind die Bedeutungen der Symbole zu entnehmen, die am Gerät angebracht sein können.

Tabelle 2: Bedeutung der am Gerät angebrachten Symbole

 Ein (Energie) IEC 417, No.5007	 Aus (Energie) IEC 417, No.5008	 Erdanschluss IEC 417, No.5017	 Schutzleiteranschluss IEC 417, No.5019
 Masseanschluss IEC 417, No.5020	 Aquipotentialanschluss IEC 417, No.5021	 Gleichstrom IEC 417, No.5031	 Wechselstrom IEC 417, No.5032
 Gleich- oder Wechselstrom IEC 417, No.5033-a	 Durchgängige doppelte oder verstärkte Isolierung IEC 417, No.5172-a	 Dreileiter-Wechselstrom (Drehstrom) IEC 617-2, No.020206	
 Warnung vor einer Gefahrenstelle (Achtung, Dokumentation beachten) ISO 3864, No.B.3.1	 Warnung vor gefährlicher elektrischer Spannung ISO 3864, No.B.3.6	 Höhere Temperatur an leicht zugänglichen Teilen IEC 417, No.5041	

Sicherheitsvorschriften und Vorsichtsmaßnahmen

Folgende allgemeine Sicherheitsvorschriften sind während allen Betriebsphasen dieses Gerätes zu befolgen. Eine Missachtung der Sicherheitsvorschriften und sonstiger Warnhinweise in dieser Betriebsanleitung verletzt die für dieses Gerät und seine Bedienung geltenden Sicherheitsstandards, und kann die Schutzvorrichtungen an diesem Gerät wirkungslos machen. MKS Instruments, Inc. haftet nicht für Missachtung dieser Sicherheitsvorschriften seitens des Kunden.

Niemals Teile austauschen oder Änderungen am Gerät vornehmen!

Ersetzen Sie keine Teile mit baugleichen oder ähnlichen Teilen, und nehmen Sie keine eigenmächtigen Änderungen am Gerät vor. Schicken Sie das Gerät zwecks Wartung und Reparatur an den MKS-Kalibrierungs- und -Kundendienst ein. Nur so wird sichergestellt, daß alle Schutzvorrichtungen voll funktionsfähig bleiben.

Wartung nur durch qualifizierte Fachleute!

Das Auswechseln von Komponenten und das Vornehmen von internen Einstellungen darf nur von qualifizierten Fachleuten durchgeführt werden, niemals vom Bedienpersonal.

Vorsicht beim Arbeiten mit gefährlichen Stoffen!

Wenn gefährliche Stoffe verwendet werden, muß der Bediener die entsprechenden Sicherheitsvorschriften genauestens einhalten, das Gerät, falls erforderlich, vollständig spülen, sowie sicherstellen, daß der Gefahrstoff die von ihm benetzten, am Gerät verwendeten Materialien, insbesondere Dichtungen, nicht angreift.

Spülen des Gerätes mit Gas!

Nach dem Installieren oder vor dem Ausbau aus einem System muß das Gerät unter Einsatz eines reinen Trockengases vollständig gespült werden, um alle Rückstände des Vorgängermediums zu entfernen.

Anweisungen zum Spülen des Gerätes

Das Gerät darf nur unter einer Ablufthaube gespült werden. Schutzhandschuhe sind zu tragen.

Gerät nicht zusammen mit explosiven Stoffen, Gasen oder Dämpfen benutzen!

Um der Gefahr einer Explosion vorzubeugen, darf dieses Gerät niemals zusammen mit (oder in der Nähe von) explosiven Stoffen aller Art eingesetzt werden, sofern es nicht ausdrücklich für diesen Zweck zugelassen ist.

Anweisungen zum Installieren der Armaturen!

Alle Anschlußstücke und Armaturenteile müssen mit der Gerätespezifikation übereinstimmen, und mit dem geplanten Einsatz des Gerätes kompatibel sein. Der Einbau, insbesondere das Anziehen und Abdichten, muß gemäß den Anweisungen des Herstellers vorgenommen werden.

Verbindungen auf Undichtigkeiten prüfen!

Überprüfen Sie sorgfältig alle Verbindungen der Vakuumkomponenten auf undichte Stellen.

Gerät nur unter zulässigen Anschlußdrücken betreiben!

Betreiben Sie das Gerät niemals unter Drücken, die den maximal zulässigen Druck (siehe Produktspezifikationen) übersteigen.

Geeignete Berstscheibe installieren!

Wenn mit einer unter Druck stehenden Gasquelle gearbeitet wird, sollte eine geeignete Berstscheibe in das Vakuumsystem installiert werden, um eine Explosionsgefahr aufgrund von steigendem Systemdruck zu vermeiden.

Verunreinigungen im Gerät vermeiden!

Stellen Sie sicher, daß Verunreinigungen jeglicher Art weder vor dem Einsatz noch während des Betriebs in das Instrumenteninnere gelangen können. Staub- und Schmutzpartikel, Glassplitter oder Metallspäne können das Gerät dauerhaft beschädigen oder Prozeß und Meßwerte verfälschen.

Geräteeinheit auf Arbeitstemperatur bringen!

Wird das Gerät zur Flußregelung gefährlicher Gase verwendet, so dürfen diese nur nach Abschluß des Anwärmvorgangs zugeführt werden. Um das versehentliche Fließen von Gas während der Aufheizperiode zu verhindern, sollte ein Absperrventil (normal geschlossen) eingebaut werden.

Informations relatives à la sécurité pour le contrôleur de débit de masse

Symboles utilisés dans ce manuel d'utilisation

Définitions des indications AVERTISSEMENT, ATTENTION, et REMARQUE utilisées dans ce manuel.



Avertissement

L'indication **AVERTISSEMENT** signale un danger pour le personnel. Elle attire l'attention sur une procédure, une pratique, une condition, ou toute autre situation présentant un risque d'accident pour le personnel, en cas d'exécution incorrecte ou de non-respect des consignes.



Attention

L'indication **ATTENTION** signale un danger pour l'appareil. Elle attire l'attention sur une procédure d'exploitation, une pratique, ou toute autre situation, présentant un risque de dégât ou de destruction partielle ou totale du produit, en cas d'exécution incorrecte ou de non-respect des consignes.



Remarque

L'indication **REMARQUE** signale une information importante. Elle attire l'attention sur une procédure, une pratique, une condition, ou toute autre situation, présentant un intérêt particulier.

Symboles figurant sur l'unité

Le tableau suivant décrit les symboles pouvant apparaître sur l'unité.

Tableau 3: Définition des symboles sur l'unité

 Marche (sous tension) IEC 417, No.5007	 Arrêt (hors tension) IEC 417, No.5008	 Terre (masse) IEC 417, No.5017	 Terre de protection (masse) IEC 417, No.5019
 Masse IEC 417, No.5020	 Equipotentialité IEC 417, No.5021	 Courant continu IEC 417, No.5031	 Courant alternatif IEC 417, No.5032
 Courant continu et alternatif IEC 417, No.5033-a	 Matériel de classe II IEC 417, No.5172-a	 Courant alternatif triphasé IEC 617-2, No.020206	
 Attention : se reporter à la documentation ISO 3864, No.B.3.1	 Attention : risque de choc électrique ISO 3864, No.B.3.6	 Attention : surface brûlante IEC 417, No.5041	

Mesures de sécurité et précautions

Observer les précautions générales de sécurité suivantes pendant toutes les phases d'exploitation de cet appareil. Le non-respect des ces précautions ou des avertissements du manuel constitue une violation des normes de sécurité relatives à l'utilisation de l'appareil et peut compromettre la protection assurée par l'appareil. MKS Instruments, Inc. rejette toute responsabilité en cas de non-respect des consignes par les clients.

PAS DE SUBSTITUTION DE PIÈCES OU DE MODIFICATION DE L'APPAREIL

Ne pas installer des pièces de substitution ou effectuer des modifications non autorisées sur l'appareil. Renvoyer l'appareil à un centre de service et de calibrage MKS pour tout dépannage ou réparation afin de garantir l'intégrité des dispositifs de sécurité.

DÉPANNAGE UNIQUEMENT PAR DU PERSONNEL QUALIFIÉ

Le personnel d'exploitation ne doit pas essayer de remplacer des composants ou de faire des réglages internes. Tout dépannage doit être uniquement effectué par du personnel qualifié.

PRÉCAUTION EN CAS D'UTILISATION AVEC DES PRODUITS DANGEREUX

Si des produits dangereux sont utilisés, prendre les mesures de précaution appropriées, purger complètement l'appareil quand cela est nécessaire, et s'assurer que les produits utilisés sont compatibles avec les composants liquides de l'appareil, y compris les matériaux d'étanchéité.

PURGE DE L'APPAREIL

Après l'installation de l'unité, ou avant son enlèvement d'un système, purger l'unité complètement avec un gaz propre et sec afin d'éliminer toute trace du produit de flux utilisé précédemment.

UTILISATION DES PROCÉDURES APPROPRIÉES POUR LA PURGE

Cet appareil doit être purgé sous une hotte de ventilation, et il faut porter des gants de protection.

PAS D'EXPLOITATION DANS UN ENVIRONNEMENT EXPLOSIF

Pour éviter toute explosion, ne pas utiliser cet appareil dans un environnement explosif, sauf en cas d'homologation spécifique pour une telle exploitation.

UTILISATION D'ÉQUIPEMENTS APPROPRIÉS ET PROCÉDURES DE SERRAGE

Tous les équipements de l'appareil doivent être cohérents avec ses spécifications, et compatibles avec l'utilisation prévue de l'appareil. Assembler et serrer les équipements conformément aux directives du fabricant.

VÉRIFICATION DE L'ÉTANCHÉITÉ DES CONNEXIONS

Vérifier attentivement toutes les connexions des composants pour le vide afin de garantir l'étanchéité de l'installation.

EXPLOITATION AVEC DES PRESSIONS D'ENTRÉE NON DANGEREUSES

Ne jamais utiliser des pressions supérieures à la pression nominale maximum (se reporter aux spécifications de l'unité pour la pression maximum admissible).

INSTALLATION D'UN DISQUE D'ÉCHAPPEMENT ADAPTÉ

En cas d'exploitation avec une source de gaz pressurisé, installer un disque d'échappement adapté dans le système à vide afin d'éviter une explosion du système en cas d'augmentation de la pression.

MAINTIEN DE L'UNITÉ À L'ABRI DES CONTAMINATIONS

Ne pas laisser des produits contaminants pénétrer dans l'unité avant ou pendant l'utilisation. Des produits contaminants tels que des poussières et des fragments de tissu, de glace et de métal peuvent endommager l'unité d'une manière permanente ou contaminer le processus.

RESPECT DU TEMPS D'ÉCHAUFFEMENT

Si l'unité est utilisée pour contrôler des gaz dangereux, ceux-ci ne doivent pas être appliqués avant l'échauffement complet de l'unité. Utiliser une valve de fermeture positive afin de garantir qu'aucun flux ne se produise par erreur pendant l'échauffement.

Medidas de seguridad del controlador de flujo de masa

Símbolos usados en este manual de instrucciones

Definiciones de los mensajes de advertencia, precaución y de las notas usados en el manual.



Advertencia

El símbolo de advertencia indica la posibilidad de que se produzcan daños personales. Pone de relieve un procedimiento, práctica, estado, etc. que en caso de no realizarse o cumplirse correctamente puede causar daños personales.



Precaución

El símbolo de precaución indica la posibilidad de producir daños al equipo. Pone de relieve un procedimiento operativo, práctica, etc. que en caso de no realizarse o cumplirse correctamente puede causar daños o la destrucción total o parcial del equipo.



Nota

El símbolo de notas indica información de importancia. Este símbolo pone de relieve un procedimiento, práctica o condición cuyo conocimiento es esencial destacar.

Símbolos hallados en la unidad

La tabla siguiente contiene los símbolos que puede hallar en la unidad.

Tabla 4: Definición de los símbolos hallados en la unidad

 Encendido (alimentación eléctrica) IEC 417, N°5007	 Apagado (alimentación eléctrica) IEC 417, N°5008	 Puesta a tierra IEC 417, N°5017	 Protección a tierra IEC 417, N°5019
 Caja o chasis IEC 417, N°5020	 Equipotencialidad IEC 417, N°5021	 Corriente continua IEC 417, N°5031	 Corriente alterna IEC 417, N°5032
 Corriente continua y alterna IEC 417, N°5033-a	 Equipo de clase II IEC 417, N°5172-a	 Corriente alterna trifásica IEC 617-2, N°020206	
 Precaución. Consulte los documentos adjuntos ISO 3864, N° B.3.1	 Precaución. Riesgo de descarga eléctrica ISO 3864, N° B.3.6	 Precaución. Superficie caliente IEC 417, N° 5041	

Procedimientos y precauciones de seguridad

Las medidas generales de seguridad descritas a continuación deben observarse durante todas las etapas de funcionamiento del instrumento. La falta de cumplimiento de dichas medidas de seguridad o de las advertencias específicas a las que se hace referencia en otras partes de este manual, constituye una violación de las normas de seguridad establecidas para el uso previsto del instrumento y podría anular la protección proporcionada por el equipo. Si el cliente no cumple dichas precauciones y advertencias, MKS Instruments, Inc. no asume responsabilidad legal alguna.

NO UTILICE PIEZAS NO ORIGINALES O MODIFIQUE EL INSTRUMENTO

No instale piezas que no sean originales o modifique el instrumento sin autorización. Para asegurar el correcto funcionamiento de todos los dispositivos de seguridad, envíe el instrumento al Centro de servicio y calibración de MKS toda vez que sea necesario repararlo o efectuar tareas de mantenimiento.

LAS REPARACIONES DEBEN SER EFECTUADAS ÚNICAMENTE POR TÉCNICOS AUTORIZADOS

Los operarios no deben intentar reemplazar los componentes o realizar tareas de ajuste en el interior del instrumento. Las tareas de mantenimiento o reparación deben ser realizadas únicamente por personal autorizado.

TENGA CUIDADO CUANDO TRABAJE CON MATERIALES TÓXICOS

Cuando se utilicen materiales tóxicos, es responsabilidad de los operarios cumplir las medidas de seguridad correspondientes, purgar totalmente el instrumento cuando sea necesario y comprobar que el material utilizado sea compatible con los materiales humedecidos de este producto e inclusive, con los materiales de sellado.

PURGUE EL INSTRUMENTO

Una vez instalada la unidad o antes de retirarla del sistema, purgue completamente la unidad con gas limpio y seco para eliminar todo resto de la sustancia líquida empleada anteriormente.

USE PROCEDIMIENTOS ADECUADOS PARA REALIZAR LA PURGA

El instrumento debe purgarse debajo de una campana de ventilación y deben utilizarse guantes protectores.

NO HAGA FUNCIONAR ESTE INSTRUMENTO EN UN AMBIENTE CON RIESGO DE EXPLOSIONES

Para evitar que se produzcan explosiones, no haga funcionar este producto en un ambiente con riesgo de explosiones, excepto cuando el mismo haya sido certificado específicamente para tal uso.

USE ACCESORIOS ADECUADOS Y REALICE CORRECTAMENTE LOS PROCEDIMIENTOS DE AJUSTE

Todos los accesorios del instrumento deben cumplir las especificaciones del mismo y ser compatibles con el uso que se debe dar al instrumento. Arme y ajuste los accesorios de acuerdo con las instrucciones del fabricante.

COMPRUEBE QUE LAS CONEXIONES SEAN A PRUEBA DE FUGAS

Inspeccione cuidadosamente las conexiones de los componentes de vacío para comprobar que hayan sido instalados a prueba de fugas.

HAGA FUNCIONAR EL INSTRUMENTO CON PRESIONES DE ENTRADA SEGURAS

No haga funcionar nunca el instrumento con presiones superiores a la máxima presión nominal (en las especificaciones del instrumento hallará la presión máxima permitida).

INSTALE UNA CÁPSULA DE SEGURIDAD ADECUADA

Cuando el instrumento funcione con una fuente de gas presurizado, instale una cápsula de seguridad adecuada en el sistema de vacío para evitar que se produzcan explosiones cuando suba la presión del sistema.

MANTENGA LA UNIDAD LIBRE DE CONTAMINANTES

No permita el ingreso de contaminantes en la unidad antes o durante su uso. Los productos contaminantes tales como polvo, suciedad, pelusa, lascas de vidrio o virutas de metal pueden dañar irreparablemente la unidad o contaminar el proceso.

PERMITA QUE LA UNIDAD SE CALIENTE

Si se utiliza la unidad para controlar gases peligrosos, no libere los gases hasta que la unidad termine de calentarse. Use una válvula de cierre positivo para impedir todo flujo no deseado durante el período de calentamiento.

Chapter One: General Information

Introduction

In many processes, semiconductor processing in particular, it is necessary to deliver different flow rates of gas mixtures to different chambers in a tool or different areas of a process chamber. Historically this would require duplicate gas delivery panels. This solution is very costly to implement, maintain, and qualify. Due to the typical operating errors of commercially available Mass Flow Controllers (MFCs) the two gas panels will not provide identical performance. Particularly in the case of supplying a single process chamber from two or more gas sources, the error in gas mixtures can become an uncontrollable variable.

In order to simplify the process tool, reduce cost, and improve control it is desirable to design one gas panel to control the process gas mixture and a device to accurately control the ratio of flow into multiple channels.

The MKS Delta™ II Flow Ratio Controller is a compact unit, which accurately divides the mass flow of gas or any mixture of gasses (Q_{in}) into two proportionate flows (Q_1 & Q_2). The flow outputs are controlled by setting the ratio (α) of the two outputs ($\alpha = Q_2/Q_1$). A ratio value may be equal to or greater than 1 to a maximum of 20. The default ratio (α) of Q_2/Q_1 can be switched via Devicenet or the on-board user interface (UI) to provide Q_1/Q_2 ratios (≥ 1).

The MKS Delta II Flow Ratio Controller uses proven thermal sensor technology to measure the N_2 equivalent flows through the unit and provides quick response to changes in ratio setpoint. The ratio of the flow outputs of the MKS Delta II Flow Ratio Controller remains constant through input flow changes or perturbations. Refer to the figure below for a block diagram of the MKS Delta II Flow Ratio Controller.

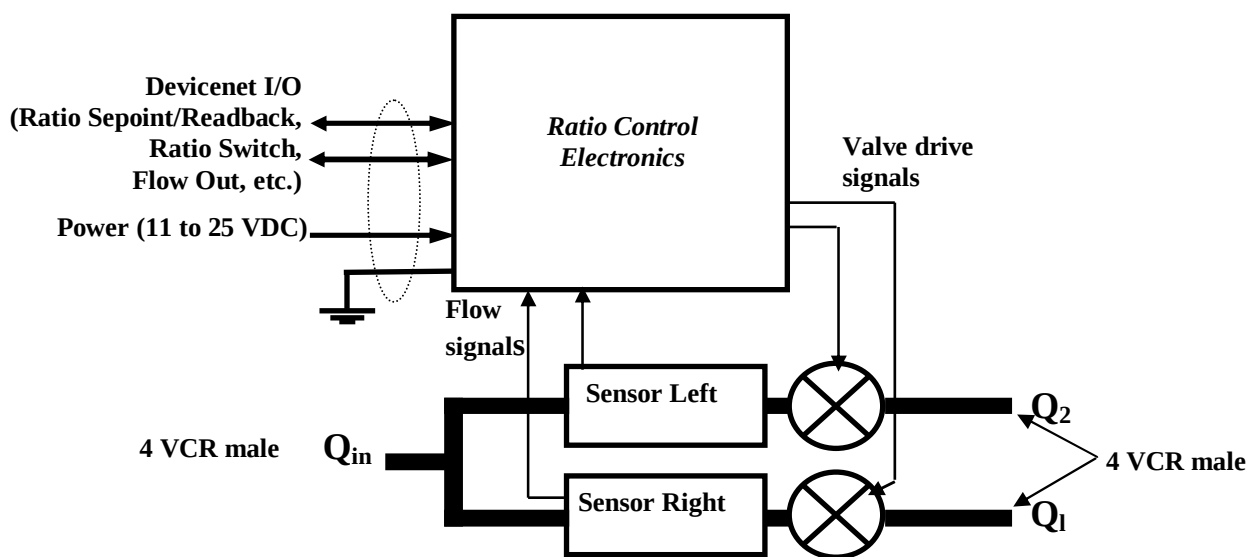


Figure 1: DELTA II Functional Block Diagram

The MKS Delta II Ratio controller contains normally open valves and is available in a variety of full-scale flow ranges to suit specific needs. A summary of some of the MKS Delta II Flow Ratio Controller properties follows:

- Ranges (N_2 Equivalent Full-scale each flow channel): 500/500, 1K/1K, 2K/2K, 10K/10K
- Connectors: Devicenet 5 pin microconnector; Ethernet RJ-45 for setup and diagnostics
- Ratio α control setpoint of 1 to 20.

- Fittings (one inlet, two outlets): Swagelok® 4-VCR® male compatible, non-rotatable
- Seals: Teflon valve plug, Internal to External: Inconel

Design Features

The design of the MKS Delta II Flow Ratio Controller incorporates two closely matched thermal mass flow sensors used to provide accurate ratiometric flow information to the control system. The control system applies control factors of a patented control algorithm directly on the flow ratio (α) independent of the actual flow through the device. As a result, the MKS Delta II Flow Ratio Controller is capable of providing ratio control where an unknown gas mixture needs to be delivered in fixed proportions to two different locations. The MKS Delta II Flow Ratio Controller provides α (Q2/Q1) proportions to be varied from 1:1 to 20:1 and, by using the ratio switch logic, α (Q1/Q2) proportions from 1:1 to 20:1. This provides an aggregate ratio control from 20:1 through 1:20. Note: This ratio switch logic is accessible via either Devicenet or Ethernet connection with a PC and an HTML browser. Solenoid-actuated proportional control valves are used and work in concert to provide the required ratio split.



Caution

The control valves within the MKS Delta II Flow Ratio Controller are not positive shutoff valves. You may need to install separate positive shutoff valves in your system.

Flow Path

Upon entering the MKS Delta II Flow Ratio Controller, the gas stream splits into two flows and enters through the metering section of each channel. The flow sensors measure the differential flow between the channels and the proportional control valves are set according to the desired ratio setpoint, ratio switch signal setting and the results of the flow sensor measurements. The gas exits the instrument at the outlet ports in the established ratio conditions.

The metering section consists of a sensor tube and parallel bypass sized for the appropriate application. The geometry of the sensor tube, in conjunction with the specified full-scale flow rate, ensures fully developed laminar flow in the sensing region. The bypass elements are specifically matched to the characteristics of the sensor tube to achieve a laminar flow splitting ratio, which remains constant throughout each range.

Ratio Flow Measurement

The flow measurement for each channel is based on differential heat transfer between two temperature sensing heater elements attached symmetrically to the sensor tube. This senses the thermal mass movement that is converted to mass flow via the specific heat, C_p , of the gas. For gases other than Nitrogen and for gas mixtures, the measured flow is the Nitrogen equivalent. The resulting signal, which is proportional to Nitrogen equivalent mass flow, is amplified and provided to the control system for operation. It is this signal that is matched closely between the two sensors inside the MKS Delta II Flow Ratio Controller.

The MKS Delta II Flow Ratio Controller uses the relative difference in voltage output between the two closely matched flow sensors to derive a ratio value rather than relying on absolute flow through each path. This method cancels the need for gas or gas mixture specificity making the knowledge of gas types and calibration for specific gas species unnecessary. Since both sensors react the same way to the gas (or gas mixture) and operate at the same input pressures, no correction factors are needed to perform accurate ratio control.

Control Range

The MKS Delta II Flow Ratio Controller is specified to control ratios of from 1:1 through 20:1. The user selects either the Q_2 or Q_1 channel to be “control” channel. The unit will control the ratio to within 2% of setpoint as long as the flow through each channel is within a 5%-100% Channel FS range (Nitrogen equivalent). The unit may operate outside these bounds, but accuracy and control are not qualified nor warranted.

Control Circuitry

The controller employs the above measurement technique and utilizes a control circuit that provides drive current for the proportioning control valves. The ratio flow controller accepts a ratio set point signal, compares it to the measured ratio output, and generates an error signal. This error signal is then conditioned by a patented control algorithm and amplified so that it repositions the control valve, thus reducing the control error to zero.

Since the proportional control valves are normally open, the MKS Delta II Flow Ratio Controller instrument closes the armature and plug assembly from the full open position to regulate the gas flow rates.

Control Valves

The control valve type used is a specially constructed solenoid valve in which the armature (moving valve mechanism) is suspended by springs. The arrangement ensures that minimal friction is present and makes precise control possible. The MKS Delta II Flow Ratio Controller has normally open valves. The control current is used to close the armature and plug assembly from the full open position, allowing a controlled flow of gas. This design maximizes system conductance and provides for a mechanism for the Delta II Flow Ratio controller to be purged without (or upon loss of) its electrical power supply.

Cleanliness Features

The design of the MKS Delta II Flow Ratio Controller ensures extremely low external leakage and minimizes a key source of particle generation, outgassing, and permeation. The design also incorporates minimal wetted surface area. To further ensure its cleanliness, the component parts of the MKS Delta II Flow Ratio Controller undergo precision machining as well as a proprietary cleaning process. The instrument is assembled in a Class 100 clean room environment and then packaged in single-bag which is then placed in a plastic case under the same Class 100 conditions.

Labels

Each MKS Delta II Flow Ratio[®] Controller carries a label that identifies the serial number and model number of the unit. The unit may also display the CE Marking which indicates compliance with European Directives.



Figure 2: Serial Number Label

How This Manual is Organized

This manual is designed to provide instructions on how to set up, install, and operate an MKS Flow Ratio Controller.

Before installing your MKS Delta II unit in a system and/or operating it, carefully read and familiarize yourself with all precautionary notes in the *Safety Information* section at the front of this manual. In addition, observe and obey all WARNING and CAUTION notes provided throughout the manual.

Chapter One, *General Information* (this chapter), introduces the product and describes the organization of the manual.

Chapter Two, *Installation*, explains environmental requirements and practical considerations to take into account when selecting the proper setting for the flow ratio controller.

Chapter Three, *Ethernet Interface Setup and Configuration*, provides information on how to setup and configure the device utilizing a PC and the instrument's internal HTML server.

Chapter Four, *Embedded Web-Based GUI and Diagnostics*, provides information on how to monitor and troubleshoot the device utilizing the instrument's internal web browser.

Chapter Five, *Operation*, describes how the flow ratio controller operates in a gas flow system. This chapter also provides information on how to override the proportional control valves within the unit. It also describes how to use a Gas Correction Factor when selecting the proper Delta II for your application.

Chapter Six, *Devicenet Protocol*, defines the application specific objects used by MKS Delta II Flow Ratio Controller and defines how system requirements are mapped to specific objects and attributes in the Devicenet protocol.

Chapter Seven, *Maintenance and Troubleshooting*, lists a few general practices to follow to ensure that the ratio flow controller will perform optimally. This chapter also includes a table of hints for reference in the event that your ratio flow controller does not perform as expected.

Appendix A, *Product Specifications*, lists the specifications of the instrument.

Appendix B, *Model Code Explanation*, describes the instrument's model code.

Appendix C, *DELTA II Sizing Guidelines*, is provided for reference and describes how to calculate the correct size Delta II for an application. This information is useful if you need to purchase another Delta II or if you plan to use your Delta II in another application different from which it was originally specified.

Appendix D, *Health and Safety Form*, provides a copy of the form at the time of this manual's release.

Customer Support

Standard maintenance and repair services are available through all of the regional MKS Calibration and Service Centers.

If any difficulties arise in the use of your device, or to obtain information about companion products MKS offers, contact any authorized MKS Calibration and Service Center. If it is necessary to return the instrument to MKS, then two actions must be completed before shipping: (1) a RMA (Return Material Authorization) number must be obtained and (2) a Health and Safety Form must be completed and included with the instrument.



Warning

All returns to MKS Instruments must be free of harmful, corrosive, radioactive, or toxic materials.

Obtaining a Return Material Authorization (RMA) Number

RMA (Return Material Authorization) numbers expedite handling and ensure proper servicing of your instrument.

RMA numbers can be obtained by contacting the MKS Calibration and Service Center or through the MKS website at: <http://www.mksinst.com/service/servicehowtoorder.aspx>.



Note Returned instruments will not be accepted without a valid RMA number displayed on the shipping container.

Health and Safety Form

A returned instrument will not be examined without a signed Health and Safety form indicating that the unit is free of harmful materials.

The Health and Safety form can be obtained on the last page of this manual or through the MKS website at: <http://www.mksinst.com/service/servicehowtoorder.aspx>.



Note Returned instruments will not be examined without a signed certificate indicating the instruments are free of harmful materials.

Please refer to the inside of the back cover of this manual for a list of MKS Calibration and Service Centers.

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Chapter Two: Installation

How to Unpack the MKS DELTA II Flow Ratio Controller Unit

MKS has carefully packed the MKS Delta II Flow Ratio Controller unit so that it will reach you in perfect operating order. Upon receiving the unit, however, you should check for defects, cracks, broken connectors, etc., to be certain that damage has not occurred during shipment.

**Note**

Do *not* discard any packing materials until you have completed your inspection and are sure the unit arrived safely.

If you find any damage, notify your carrier and MKS immediately. If it is necessary to return the unit to MKS, obtain an RMA Number (Return Material Authorization Number) from the MKS Service Center before shipping. A completed Health and Safety Form is also required for any return. Please refer to the inside back cover of this manual for a list of MKS Calibration and Service Centers.

Opening the Package

The MKS Delta II Flow Ratio Controller unit is assembled, leak tested with helium, and calibrated in a clean room environment. The instrument is single-bagged and placed in a plastic case in this environment to ensure maintenance of its particle free condition during shipment. It is very important to remove the packaging according to good clean room practices. To maintain at least a minimal level of clean room standards, follow the instructions below.

1. Remove the plastic container in an ante room (garmenting room) or transfer box.

Do not allow this plastic container to enter the clean room.

2. Wipe down the exterior of the inner bag with a clean room wipe.

This step reduces the contamination introduced into the clean room.

3. Remove the inner bag in the clean room.

**Caution**

Only qualified individuals should perform the installation and any user adjustments. Individuals must comply with all necessary ESD handling precautions while installing and adjusting the instrument. Proper handling is essential when working with all highly sensitive precision electronic instruments.

Unpacking Checklist

Standard Equipment:

- The MKS DELTA II Flow Ratio Controller
- The MKS DELTA II Flow Ratio Controller instruction manual (this book)

Environmental Requirements

Follow the guidelines listed below when installing and using the MKS DELTA II Flow Ratio Controller.

- Ventilation requirements include sufficient air circulation. Allow 2” around each surface of the device for adequate ventilation.
- Maintain the normal operating temperature between 10° and 40° C (50° and 104° F).
- Observe the device operating pressure limits:
 - A. Operational differential pressure is ≤ 150 Torr (≤ 400 Torr for 10K/10K model).
- Provide power input at +11 to 25 VDC
 - A. Maximum voltage/current at startup is +24VDC ($\pm 5\%$) @ 1A.
 - B. Typical steady state voltage/current should be +24VDC ($\pm 5\%$) @ 375 mA.
- Allow 60 minutes for warm-up time.
- Use high purity gas (minimal moisture and particulates) and gas filters in line upstream of the Delta II.
- Install separate positive shutoff valves if your system cannot tolerate leakage across the proportional control valves in the MKS Delta II Flow Ratio Controller. The proportional control valves are not positive shutoff valves so some leakage across the valves may occur even when commanded closed. In an unpowered condition, the valves are normally open.



Warning

Follow your corporate policy for handling toxic or hazardous gases. Your corporate policy on handling these gases *supersedes* the instructions in this manual. MKS assumes no liability for the safe handling of such materials.

For additional information, refer to Appendix A, *Product Specifications*, page 87.

Interface Cables

As of January 1, 1996, most products shipped to the European Community must comply with the EMC Directive 89/336/EEC, which covers radio frequency emissions and immunity tests. In addition, as of January 1, 1997, some products shipped to the European Community must also comply with the Product Safety Directive 92/59/EEC and Low Voltage Directive 73/23/EEC, which cover general safety practices for design and workmanship. MKS products that meet these requirements are identified by application of the CE Mark.

To ensure compliance with EMC Directive 89/336/EEC, an overall metal braided shielded cable, properly grounded at both ends, is required during use. No additional installation requirements are necessary to ensure compliance with Directives 92/59/EEC and 73/23/EEC.



Notes

1. An overall metal braided, shielded cable, properly grounded at both ends, is required during use to meet CE specifications.
2. Use an overall metal braided shielded cable assemblies, especially if the environment contains high EMI/RFI noise.
3. Provide adequate clearance for Type “D” cable assemblies:
 - Straight Shielded connectors require approximately 3” height.
 - Right Angle connectors require approximately 2” height.

Generic Shielded Cable Guidelines

MKS offers cables for all MKS equipment. To manufacture your own cables, follow these guidelines.

1. The cable must have an overall metal *braided* shield, covering all wires. Neither aluminum foil nor spiral shielding will be as effective—using either may nullify regulatory compliance.
2. The connectors must have a metal case with direct contact to the cable shield on the whole circumference of the cable. The inductance of a flying lead or wire from the shield to the connector will seriously degrade effectiveness of the shield. Ground the shield to the connector before its internal wires exit.
3. With few exceptions, the connector(s) must make good contact to the device case (ground). “Good contact” is about 0.01 ohms and the ground should surround all wires. Contact to ground at just one point may not suffice.
4. For shielded cables with flying leads at one or both ends, it is important to ground the shield at each such end *before* the wires exit. Make this ground with absolute minimum length. (A ¼ inch piece of #22 wire may be undesirably long since it has approximately 5 nH of inductance, equivalent to 31 ohms at 1000 MHz.) After picking up the braid ground, keep the wires and braid flat against the case. With very few exceptions, grounded metal covers are not required over terminal strips. If one is required, it will be stated in the Declaration of Conformity.
5. In selecting the appropriate type and wire size for cables, consider:
 - A. Voltage ratings.
 - B. Cumulative I^2R heating of all the conductors (keep them safely cool).
 - C. IR drop of the conductors, so that adequate power or signal voltage gets to the device.
 - D. The capacitance and inductance of cables that are handling fast signals (such as data lines or stepper motor drive cables).
 - E. That some cables may need internal shielding from specific wires to others.

Setup

Follow the guidelines below when setting up the MKS Delta II Flow Ratio Controller.

1. Set the controller into position where it will be connected to a gas supply.

Placement of flow components in an orientation other than that in which they were calibrated (horizontal, base down) may cause a small zero shift in the flow sensor and ratio output. The zero offset can be removed according to the instructions found in the Maintenance and Troubleshooting section of this manual.
2. Install the ratio flow controller in the gas stream such that the flow will be in the direction of the arrow on the side of the controller.
3. Allow adequate clearance for the Devicenet cable connector and gas line tubing.

Refer to Figures 3 through 7, pages 18 through 20 for outline dimensions and for mounting dimensions of the ratio flow controller.

Dimensions



Note All dimensions are listed in inches (mm).

Inlet Side View

The inlet side of the MKS Delta II Flow Ratio Controller has the 5 pin male microconnector for power and Devicenet communications with pinouts silkscreened on the enclosure, a RJ-45 female connector for access to the internal setup and diagnostics capability via an HTML based server, a Swagelok 4-VCR non-rotatable male compatible inlet fitting.

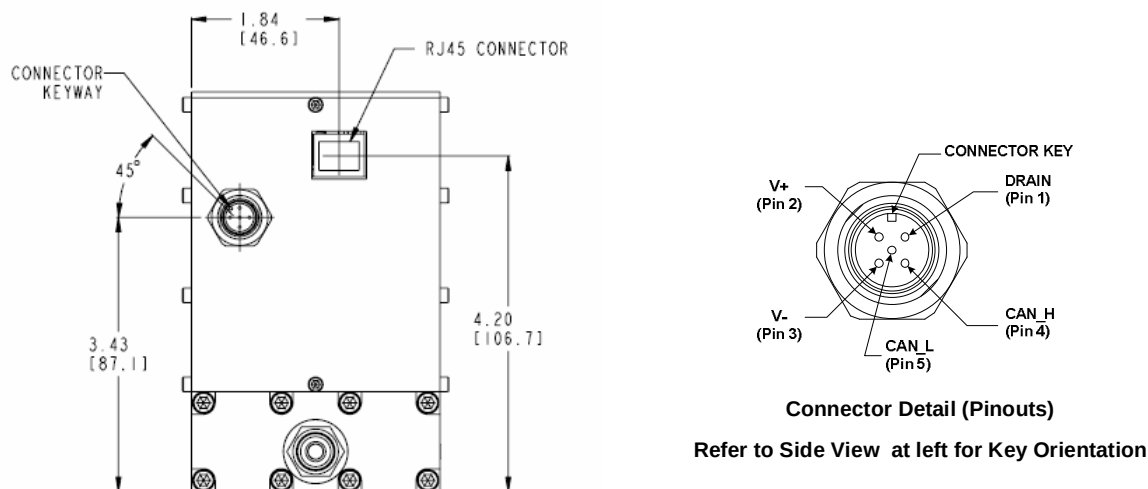


Figure 3: Inlet Side View of the MKS DELTA II Flow Ratio Controller (Devicenet)

Top View

The top of the MKS Delta II Flow Ratio Controller has a label identifying the model and serial number of the Delta, Devicenet MACID and baud rate switches, Net and Mod status LEDs, and a four digit LED which displays ratio output 30 seconds after power is applied to the device.

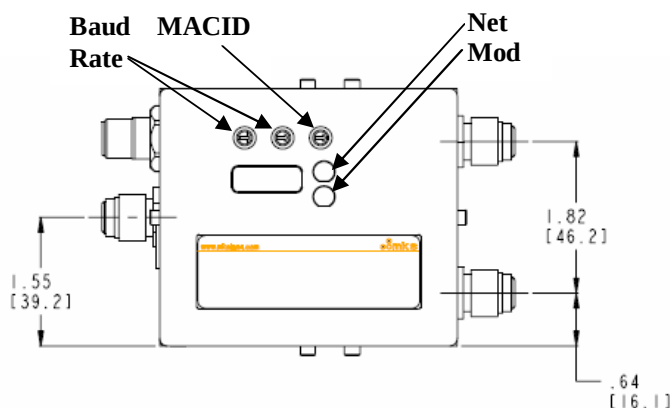


Figure 4: Top View of the MKS DELTA II Flow Ratio Controller (Devicenet)

Bottom View (Mount)

The bottom view of the MKS Delta II Flow Ratio Controller shows the location and size of the mounting holes for the instrument.

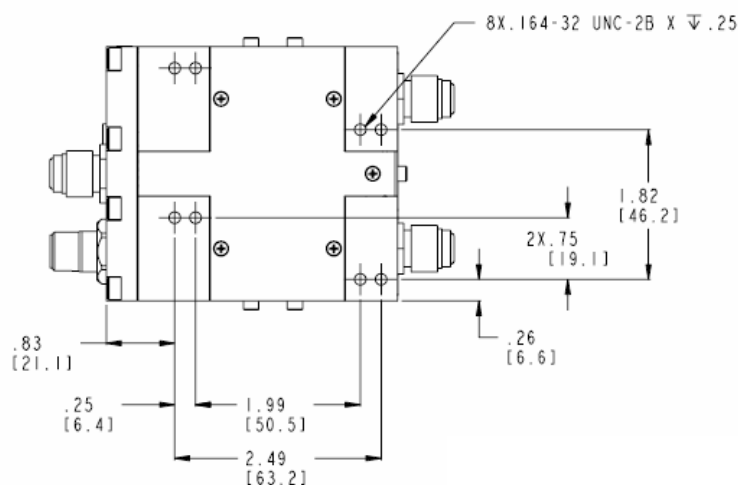


Figure 5: Bottom View of the MKS DELTA II Flow Ratio Controller (Devicenet)

Front View

The front view of the MKS Delta II Flow Ratio Controller has the device name, MKS logo, and indicates the gas flow direction. It also shows the inlet and outlet (Swagelock 4-VCR male) fittings.

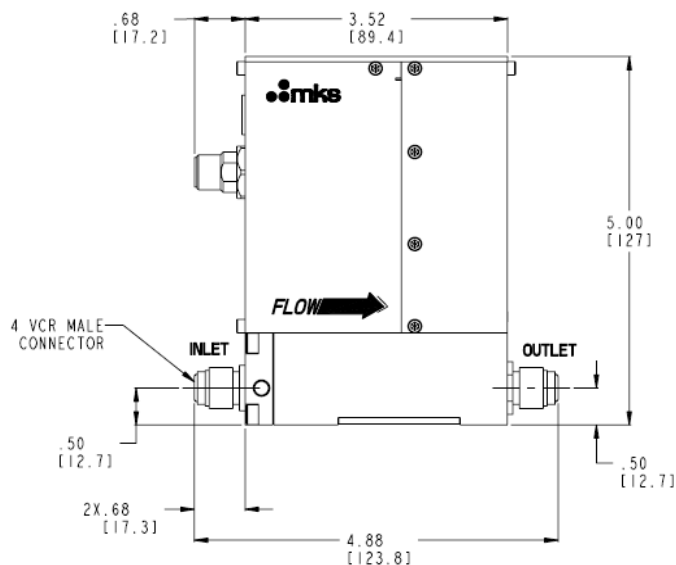


Figure 6: Front View of the MKS DELTA II Flow Ratio Controller (Devicenet)

Outlet End View

The outlet end view of the MKS Delta II Flow Ratio Controller shows both channel outlet 4-VCR non-rotatable male fittings. Notice that flow 1 (Q1) is on the left when viewed from the outlet end.

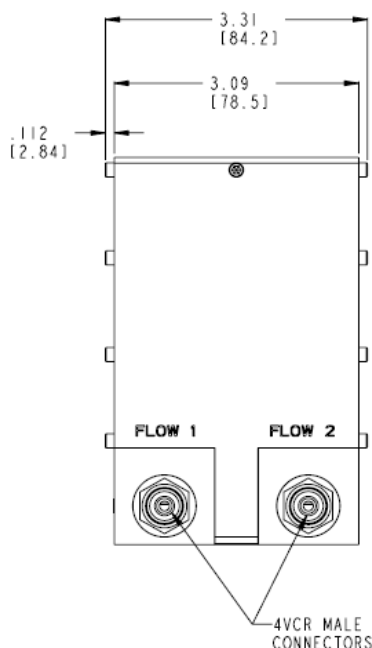


Figure 7: Outlet Side View of MKS DELTA II Flow Ratio Controller (Devicenet)

Fittings

The MKS Delta II Flow Ratio Controller is equipped with Swagelok compatible 4-VCR male non-rotatable fittings. For specific information regarding these fittings, refer to the manufacturer's documentation.

Mounting an MKS DELTA II Flow Ratio Controller

The blind tapped holes located on the bottom of the Delta II are used for mounting the unit. The system to which the Delta II is mounted should use slotted clearance openings to allow for system and Delta II tolerances. Refer to Figure 5, page 19, for the size and location of the mounting holes on the bottom surface of the Delta II.

Control Valves

The normally open proportional control valves, housed inside the MKS Delta II Flow Ratio Controller enclosure, are *not* positive shutoff valves. Some leakage across the valves may occur. Refer to Appendix A, *Product Specifications*, page 87, for the closed conductance and external leak integrity specifications. If necessary, install separate positive shutoff valves in your system.

Installation Procedure

Install the Delta II Flow Ratio Controller


Note

DO NOT make any electrical connections to the Delta II Ratio Controller until directed to do so.

1. The Delta II Ratio Controller is prepared for cleanroom installation. Follow standard cleanroom practices to ensure a clean installation:
 - Discard outer material outside of the cleanroom.
 - Remove the outer packaging in the gray area and wipe down inner bag.
 - Carry the Delta II Ratio Controller into the clean area then remove the inner bag and any protective fitting covers just prior to installation.
 - Do not discard calibration sheet.
2. Prepare the system into which the Delta II Ratio Controller will be installed according to your facility's gas handling procedures, including purging of the gas lines with appropriate purge gas, and notification to equipment personnel, safety engineers, environmental health and safety management, and haz/mat response teams.


PERSONAL SAFETY HAZARDS!

Gas systems can contain toxic, explosive, combustible, corrosive or other gases which can present life-threatening hazards. **ALWAYS** use appropriate personal protection equipment. **NEVER** open a gas line unless the system has been properly purged of harmful gases. Certain gas system components may contain hazardous residuals if not properly prepared. Consult with your facility safety engineers prior to working on any gas delivery system and notify all personnel in adjacent areas to take appropriate personal safety precautions **BEFORE** working on the equipment.

3. Prepare the Delta II Ratio Controller inlet and outlet gas connections:
 - Flow clean, dry purge gas across the fittings to minimize particle contamination during installation. Use only purge gases that are approved for your process and compatible with the wetted materials of the Delta II Ratio Controller.
 - Install the Delta II Ratio Controller and secure it to the system using the appropriate mounting hardware.
 - Secure the inlet and outlet fitting connections according to the fitting manufacturer's instructions. **DO NOT** over tighten connections.


Caution

Do not overtighten the Delta II Ratio Controller VCR connections as this can result in damage to the VCR bead surface resulting in leaks.


Caution

Do not grasp the Delta II Ratio Controller electronics enclosure to provide torque resistance while tightening the VCR connections as this can result in damage to the device. Use a backing wrench when tightening these connections.

4. Verify the power and signal pinouts for the cable leading to the Delta II Flow Ratio Controller. See Connector Detail in Figure 3 for 5 pin microconnector pinouts.

When using the built-in Ethernet interface for setup, an unshielded twisted pair CAT 5E with RJ-45 males connectors wired in crossover fashion is required when the Delta II Flow Ratio Controller is connected directly to a PC's Ethernet port. When the Delta II Ratio Controller is on a network with a hub interface or switch, a standard Ethernet cable should be used.

5. Connect the 5 pin microconnector to the Delta II and power up the Delta II Ratio Controller, verify that the Maximum Flow Rate (Full Scale Flow Rate) match those on the calibration report sheet.

On power up, the Delta II Ratio Controller's scrolling display shows the IP address, the Gas Type, and the Full Scale Flow Rate for both Q_1 and Q_2 . After 30 seconds the display switches to the measured ration output unless otherwise altered through a change in configuration through the Graphical User Interface. To access other information utilize the UI's Device Page in the Setup Mode, See Chapter Four, page 35, for how to use the embedded user interface.



Note When there is an active Devicenet connection to the Delta II Flow Ratio Controller, the Ethernet UI may not be used to change any device parameters.

6. Perform appropriate helium leak checking of your gas lines and Delta II Flow Ratio Controller gas line connections to verify the integrity of the gas seals. You will need to open/close appropriate pneumatic isolation valves and proportional control valves as needed to isolate and identify any leaks in the system.

Chapter Three: Ethernet Interface Setup and Configuration

The Ethernet interface is a supplemental interface that can be used for Delta II Ratio Controller setup, configuration, and diagnostics purposes. It is not used to control the Delta II Ratio Controller during normal operation. To access the diagnostic features of the Delta II Ratio Controller via the Ethernet port, follow Steps 1 and 2.

Step 1: Install the Java™ Plug-in (for single IP address)

The Delta II Ratio Controller interface software uses a web-based Internet Explorer interface that requires a Java Technology plug-in to display real-time data plots.

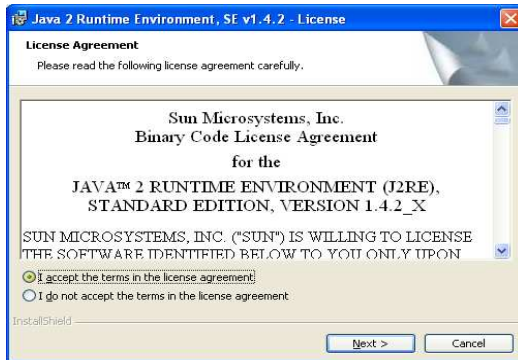
If you are installing the Delta II Ratio Controller on a network that has web access AND you are setting up multiple IP addresses, then go to *Option 3: For Multiple IP Address Setup* on page 28 and skip the steps below. The “Multiple IP Address Setup” procedure enables you to access the web for download at completion.

OTHERWISE perform the following steps:

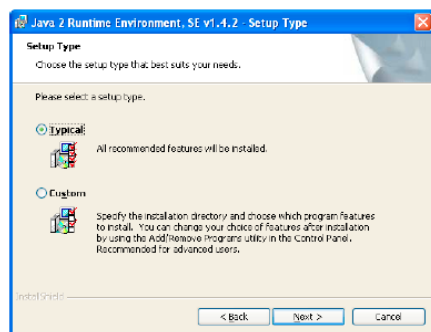
1. Download the (2) files listed below from the MKS website (www.mksinst.com/MDsw.html) by clicking on the link “Delta II Ratio Controller java plugin file for web access.” The MKS download includes an installation script to properly load the plug-in.
 - Java installer: jinstaller.exe
 - Installation script: InstallPlot.bat
2. Copy the installer and script file to your hard drive, then double-click on the InstallPlot.bat file. This file connects to the Sun Microsystems download site according to the following command:
 - `jinstall.exe http://java.sun.com/update/1.4.2/1.4.2-b28.xml`
3. Follow the onscreen prompts to install the Java application.



4. Read the license agreement, select “I accept...”, click “Next” to continue installation.



5. Select the “Typical” installation option and follow the prompts to install the Java plug-in.



6. Java plug-in is now complete.

Step 2: Setup Network for Communication through Ethernet

There are three possible ways to setup your network for communication through Ethernet. Choose the correct option based on the following criteria:

Option 1 (see below) [single port/single device]

- The Java applet discussed in Step 1 on page 23 has already been installed on your computer.
- You have no need to connect to the Internet.
- You only have one IP address, i.e. 192.168.2.XXX to connect with.

Option 2 (see page 26) [single port/multiple devices]

- The Java applet discussed in Step 1 on page 23 has already been installed on your computer.
- You have no need to connect to the Internet.
- You have multiple IP addresses you want to connect with, i.e. 10.X.X.X, 192.168.2.XXX, etc.

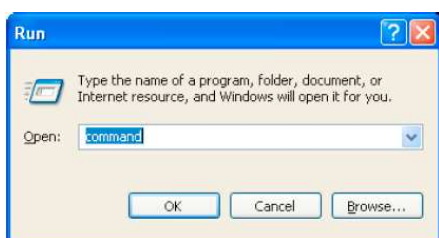
Option 3 (see page 28) [multiple ports/multiple devices]

- The Java applet discussed in Step 1 on page 23 has or has not installed on your computer.
- You need to be able connect to the Internet while connecting to one or multiple units.
- You have multiple IP addresses to connect with, i.e. 10.X.X.X, 192.168.2.X, etc.

Option 1: Network Automatic Setup

A software script allows you to rapidly create a network connection to the Delta II Ratio Controller. Once connected a series of web-browser type windows allow you to easily monitor and configure the Delta II Ratio Controller.

1. Logon to the MKS website (www.mksinst.com/MDsw.html) and download a copy of the IP setup script, HostIP.cmd by clicking on the HostIP.cmd link.
2. Copy the setup script HostIP.cmd to your C:\ directory. (Use Windows Explorer or similar program.)
3. Connect a **crossover** network cable to the Delta II Ratio Controller and your laptop computer. A crossover cable is required when the Delta II Ratio Controller is connected directly to a computer. When the Delta II Ratio Controller is connected to a network using a hub interface, a standard Ethernet cable can be used.
4. From your computer's "Start" menu, select "Run..." then enter the word "command" and click OK.



5. At the command prompt, if you find that you are not in the C-drive, then enter the command "C:". If you find yourself in a sub-directory of your C-drive, then enter the command "CD \". This will bring you to the C:\> prompt.
6. Enter the command "HostIP xx.xx.xx.xx" where the x's stand for the IP address of your Host computer, then press Enter. The system will setup a new host IP address and display it as shown in the example below:



Note

The Delta II Ratio Controller's IP Address format is 192.168.2.XXX. Your Host computer's IP Address must also be of the format 10.X.X.X. Remember the Host computer IP Address must be unique. A recommended IP Address for your Host computer is 10.0.0.X with a Subnet Mask of 255.0.0.0, where X is some number between 0 and 255.

```
C:\WINNT\system32\cmd.exe
Microsoft Windows 2000 [Version 5.00.2195]
(C) Copyright 1985-2000 Microsoft Corp.

X:\>c:

C:\>HostIP 10.0.0.1
*****
Ip Address.....: 10.0.0.1
Subnet Mask.....: 255.0.0.0
Default Gateway.....: 10.0.0.0
*****
Setting the new IP address ...
Local Area Connection is set to static
C:\>_
```

To restore to a dynamic IP address, run the HostIP script with an argument of "auto" as shown below:

```
C:\WINNT\system32\cmd.exe

C:\>HostIP auto
Setting the new IP address ...
Local Area Connection is set to dynamic
C:\>
```

These commands may also be entered directly in the “Run...” window. Instead of typing “command” simply type “c:/HostIP 10.0.0.X” (without “ ”) or “c:/HostIP auto” (without “ ”).

- After running the HostIP command, launch Internet Explorer, enter “http://xx.xx.xx.xx” in the address field, where xx.xx.xx.xx stands for the IP address of the Delta II Ratio Controller you wish to connect to, then click “Go.” Internet Explorer will open and display the Device Page in the Monitor Mode. This page will show the device full scale ranges for each flow channel, model number, and serial number. The Serial Number can always be found in the bottom left-hand corner of the browser window. See Chapter Four, page 35, for details.

For more information on the web-based program, go to Chapter Four on page 35.

Option 2: Manual Setup

If you have not setup automatic network script in Option 1, a manual setup is required to access the Delta II Ratio Controller Ethernet interface.

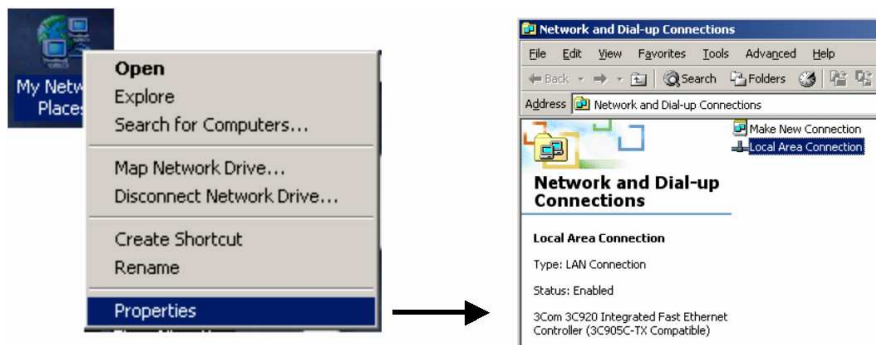
- Connect a crossover network cable to the Delta II Ratio Controller and your laptop computer.

A crossover cable is required when the Delta II Ratio Controller is connected directly to a computer. When the Delta II Ratio Controller is connected to a network using a hub interface, a standard Ethernet cable can be used.

- Open the Local Area Connection by doing one of the following options:

Option 1:

Select **My Network Places** (On Desktop typically). Right-Click on the icon, then select **Properties**.

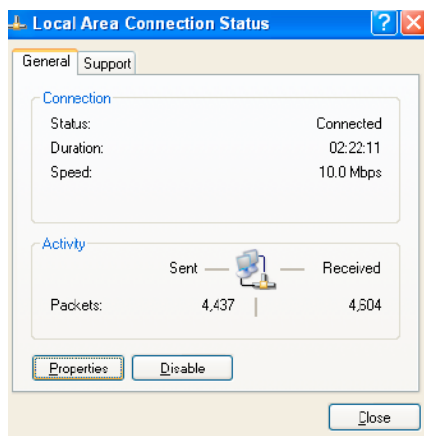


In the “Network and Dial-Up Connections” window, Double-click on “Local Area Connection.” Not all Local Area Connections have the same name, yours may have a different name.

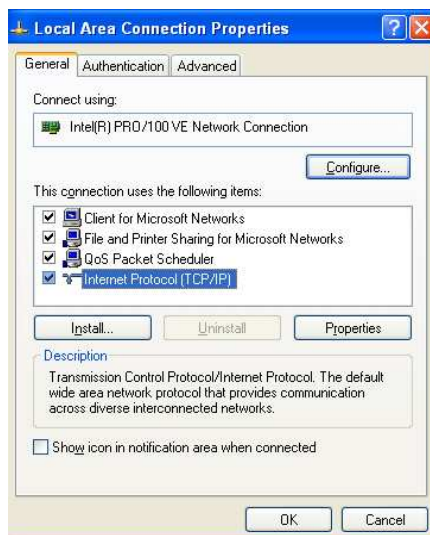
Option 2:

From your computer’s “Start” menu, select “Settings” → Network and Dialup Connections → Local Area Connections

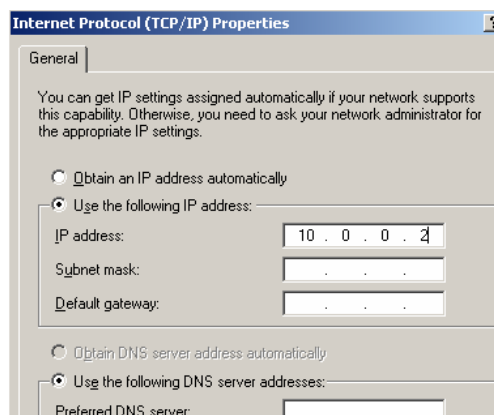
3. Select **Properties**.



4. Select **Internet Protocol (TCP/IP)**, then select **Properties**.



5. Select **Use the following IP address**.

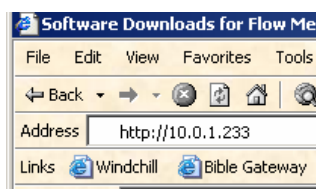


6. Enter the IP address **192.168.2.XXX**, where “X” is a number from 0 to 255 that is unique from any other IP address on your network. Now click in the Subnet mask field and the correct Subnet mask (**255.0.0.0**) should appear. If it doesn’t, you need to enter it. Also make sure that the “**Use the following DNS server Addresses**” is also selected. Leave this blank.

Additional IP Addresses

In the case that you need to be able to connect to more than one IP Address, simply click on the “Advanced” button at the bottom of the current window. Once the “Advanced TCP/IP Settings” window pops up, verify that you’re on the “IP Settings” tab and then click “Add” in the “IP Addresses” section. If you have a Delta II Ratio Controller or any of the other MKS “ π ” Series products, you may also need to add the IP Address 192.168.2.X, where “X” is a number from 0 to 255 that is unique from any other IP address on your network. Now click in the Subnet mask field and the correct Subnet mask (255.255.255.0) should appear. If it doesn’t appear, you will need to enter it.

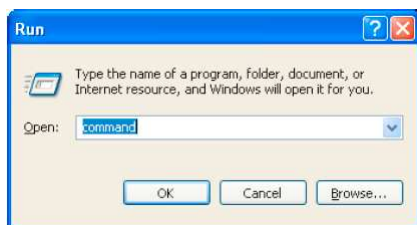
7. Close out all dialog boxes by selecting **O.K.**, **Close**, etc. as required.
8. Launch Internet Explorer (or a similar program) and enter “http://xx.xx.xx.xx” in the address field, where xx.xx.xx.xx stands for the IP address of the Delta II Ratio Controller you wish to connect to, then click “Go.” Internet Explorer will open and display the Device Page in the Monitor Mode. This page will show the device full scale ranges for each flow channel, model number, and serial number. The Serial Number can always be found in the bottom left-hand corner of the browser window. See Chapter 4 for details.



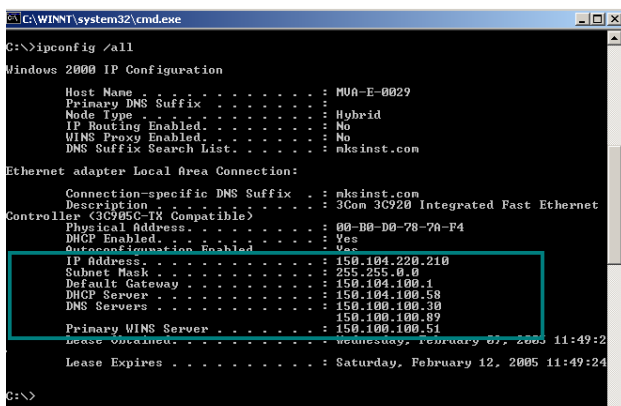
For more information on the web-based program, go to Chapter Four on page 35.

Option 3: For Multiple IP Address Setup

1. From your computer’s “Start” menu, select “Run...” then enter the word “command” and click OK.



2. At the command prompt, if you find that you are not in the C-drive, then enter the command “C:”. If you find yourself in a sub-directory of your C-drive, then enter the command “CD \”. This will bring you to the C:\> prompt.
3. Now, before going any further it is important that you have an active network connection that allows you to connect to the Internet. If you do not have an active connection, set that up and then continue.
4. At the command prompt, enter the command “ipconfig /all” (as shown below) then press Enter.



```

C:\WINNT\system32\cmd.exe
C:\>ipconfig /all

Windows 2000 IP Configuration

Host Name . . . . . : MUA-E-0029
Primary DNS Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : mksinst.com

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . : mksinst.com
Description . . . . . : 3Com 3C920 Integrated Fast Ethernet
Controller (3C905C-TX Compatible) . . . . . :
Physical Address. . . . . : 00-B0-D0-78-F4
DHCP Enabled. . . . . : Yes
Autotest Configuration Enabled . . . . . : Yes
IP Address. . . . . : 150.104.220.210
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . : 150.104.100.1
DHCP Server . . . . . : 150.104.100.58
DNS Servers . . . . . : 150.100.100.30
Primary WINS Server . . . . . : 150.100.100.89
Lease Obtained. . . . . : Wednesday, February 07, 2005 11:49:22
Lease Expires . . . . . : Saturday, February 12, 2005 11:49:24

C:\>

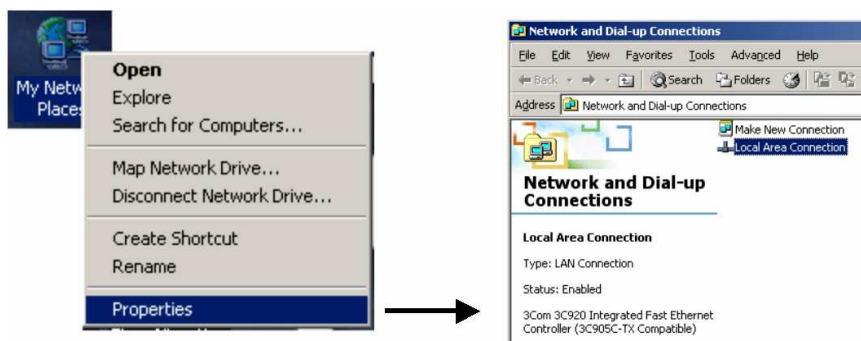
```

The data inside the highlighted area (lower section, between IP Address and Primary WINS Server) are what is needed for the steps below.

5. Open the Local Area Connection by doing one of the following options:

Option 1:

Select **My Network Places** (On Desktop typically). Right-Click on the icon, then select **Properties**.

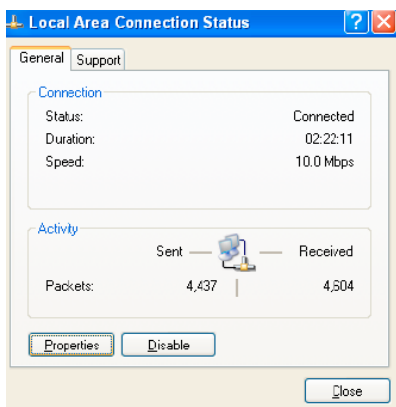


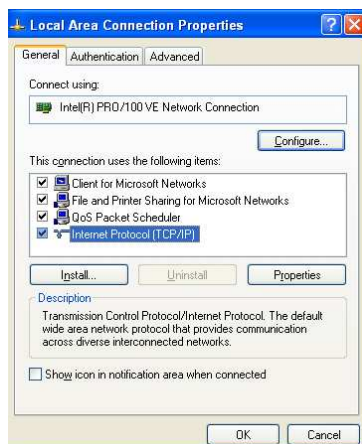
In the “Network and Dial-Up Connections” window, Double-click on “Local Area Connection.” Not all Local Area Connections have the same name, yours may have a different name.

Option 2:

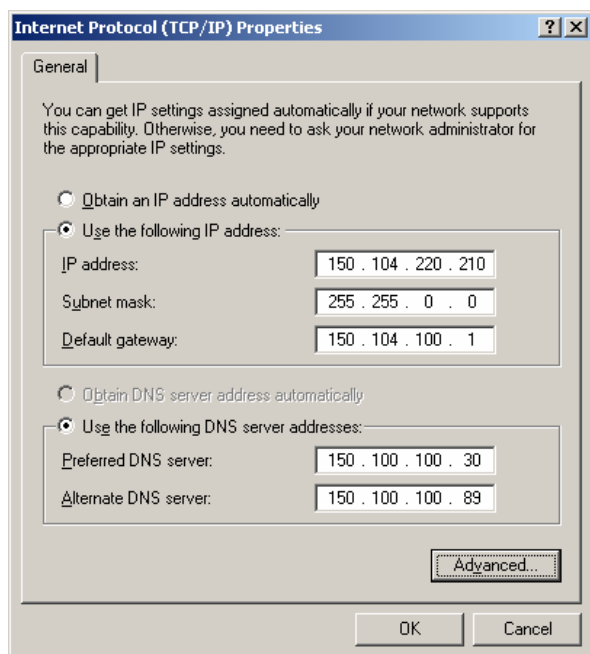
From your computer’s “Start” menu, select “Settings” → Network and Dialup Connections → Local Area Connections

6. Select **Properties**.

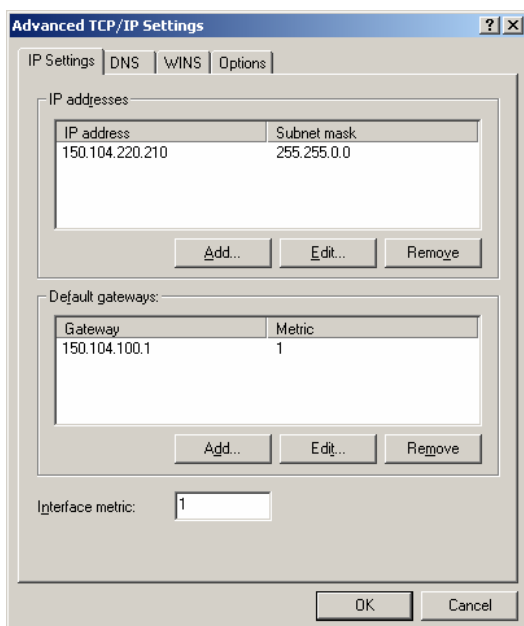


7. Select **Internet Protocol (TCP/IP)**, then select **Properties**.8. Select **Use the following IP address**. Now do the following to configure your computer so that it can still connect to the Internet:

- Type in the “IP address”, the “Subnet mask” and the “Default gateway” fields with values returned from the “ipconfig /all” query in the DOS command window.
- Select “Use the following DNS server addresses:”
- Type in the Preferred DNS server and the Alternate DNS server fields with values returned from the “ipconfig /all” query in the DOS command window. An example is shown below:

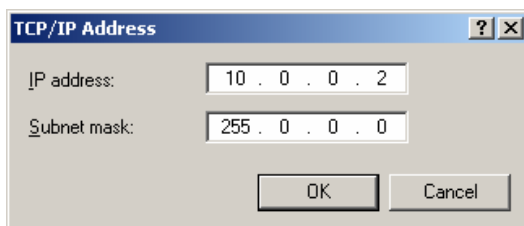


9. Click the “Advanced...” button at the bottom of the window. The following window will be displayed:

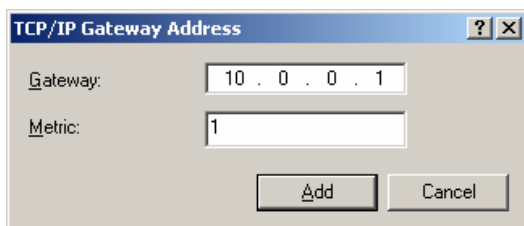


10. Verify you’re on the “IP Settings” tab and then do the following to configure your computer to talk to the Delta II Ratio Controller and any other devices with an IP address:

- Click “Add...” in the “IP address” section and enter the IP address **192.168.2XXX**, where “X” is a number from 0 to 255 that is unique from any other IP address on your network. Now click in the Subnet mask field and the correct Subnet mask (**255.0.0.0**) should appear. If it doesn’t, you need to enter it. Then click “OK.”

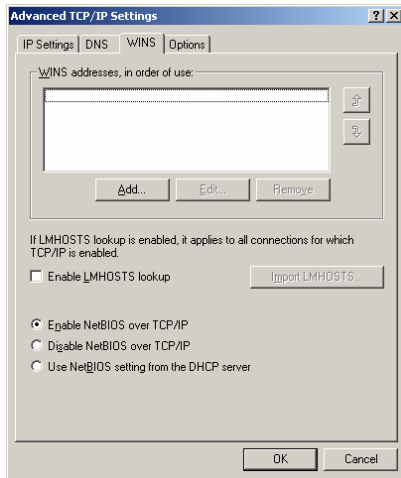


11. Now click “Add...” in the “Default gateways” group. Enter the Gateway value you will use to connect to the π Delta II Ratio Controller. Typically a gateway value will be the IP address with a (1) on the end. The Metric value should always be (1). Then click “Add.”

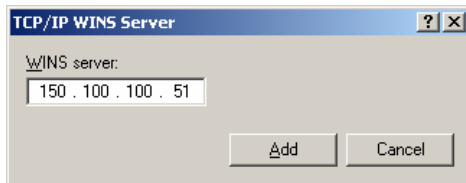


12. In the case that you need to be able to connect to more than one IP Address, i.e. not all devices on your network use a 10.X.X.X address, more addresses can be added by repeating steps 10 and 11 above. If you have a Delta II Ratio Controller or any of the other MKS “ π ” Series products, you may also need to add the IP Address 192.168.2.X, where “X” is a number from 0 to 255 that is unique from any other IP address on your network. The Subnet mask for this IP address is 255.255.255.0, and the typical Default Gateway is 192.168.2.1.

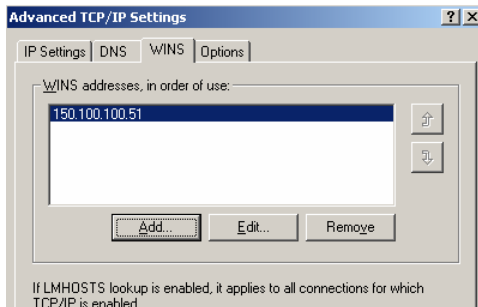
13. Click the “WINS” tab.



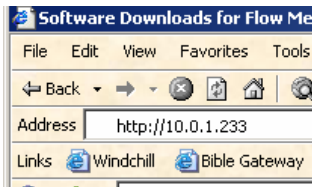
14. In the “WINS addresses, in order of use” section, click “Add...” Enter in the Primary WINS server address that was returned from the “ipconfig /all” query in the DOS command window, then click “Add.” If there were any alternate WINS servers listed in the DOS window, now enter them. If not go ahead to the next step.



Example is shown below:



15. Close out all dialog boxes by selecting **O.K.**, **Close**, etc. as required.
16. You are ready to connect to multiple IP addresses from your computer. You can connect both to the Internet and to your Delta II Ratio Controller local network.
17. Launch Internet Explorer (or a similar program) and enter “http://xx.xx.xx.xx” in the address field, where xx.xx.xx.xx stands for the IP address of the Delta II Ratio Controller you wish to connect to, then click “Go.” Internet Explorer will open and display the Device Page in the Monitor Mode. This page will show the device full scale ranges for each flow channel, model number, and serial number The Serial Number can always be found in the bottom left-hand corner of the browser window. See Chapter Four, page 35, for details.



For the plot web page to work in the Delta II Ratio Controller, a java plug-in has to be installed on your computer.

With the multiple IP addresses setup, if the Java plug-in has NOT already been installed, the IE browser will automatically connect to the correct web site for the downloading the first time someone connects to a Delta II Ratio Controller and clicks on the "Plot" page. If, for some reason, this does not work follow the Step 1 instructions (see page 23).

For more information on the web-based program, go to Chapter Four on page 35.

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Chapter Four: Embedded Web-Based GUI and Diagnostics

Logging On to Your Delta II Flow Ratio Controller

Before trying to logon to your Delta II Ratio Controller you must have your network setup correctly. To do this, complete the steps listed in Chapter Three, *Ethernet Interface Setup and Configuration*, starting on page 23.

Once you complete the steps in Chapter Three, launch Internet Explorer (or a similar program) and enter “http://xx.xx.xx.xx” in the address field, where xx.xx.xx.xx stands for the IP address of the Delta II Ratio Controller you wish to connect to, then click “Go”. Internet Explorer will open and display the Device page in Monitor Mode.

The modes are described in detail below.



Note

The Delta II Ratio Controller’s UI does not allow control of the device while there is an active Devicenet connection. To control the device using the UI the Delta II must be first disconnected from the Devicenet network. Power must still be supplied via the Devicenet connector.

Monitor Mode

Holding true to its name, “Monitor Mode” allows the user to only *Monitor* the Delta II Ratio Controller’s performance aside from performing diagnostics. In order to configure the Delta II Ratio Controller, i.e. zero the device, change the control channel, etc., you must enter “Setup” mode. See *Setup Mode*, page 41.

Each of the pages, i.e. tabs, in Monitor Mode are listed and described below in detail.

Device Page – Monitor Mode

This page, which contains the general information for the Delta II Ratio Controller, is where you are initially directed once you logon to the Delta II Ratio Controller.

What’s Displayed: Delta II Ratio Controller

The Device Page displays the device’s channel full scales flows in sccm (Nitrogen equivalent) allowed for the device.



Note

The Delta II Ratio Controller’s **serial number** can always be found in the bottom left-center of the browser window. Immediately to the right of the serial number is the **model (part) number** for the Delta II as it was configured when it left the factory.

Figure 8, next page, shows a screen capture of the Device Page for the Delta II Ratio Controller.

Device Settings: Under Device Settings are the Full Scale ranges of Channel 1 (Q₁) and Channel 2 (Q₂). These full scale ranges are given in terms of Nitrogen equivalent gas flow.

Firmware and

Calibration Date: **Firmware version:** This is the version of the firmware that is implemented on the device.
Devicenet version: This is the version of the Devicenet code that is implemented on the device.

Calibration date: This is the date that the Delta II was last calibrated.

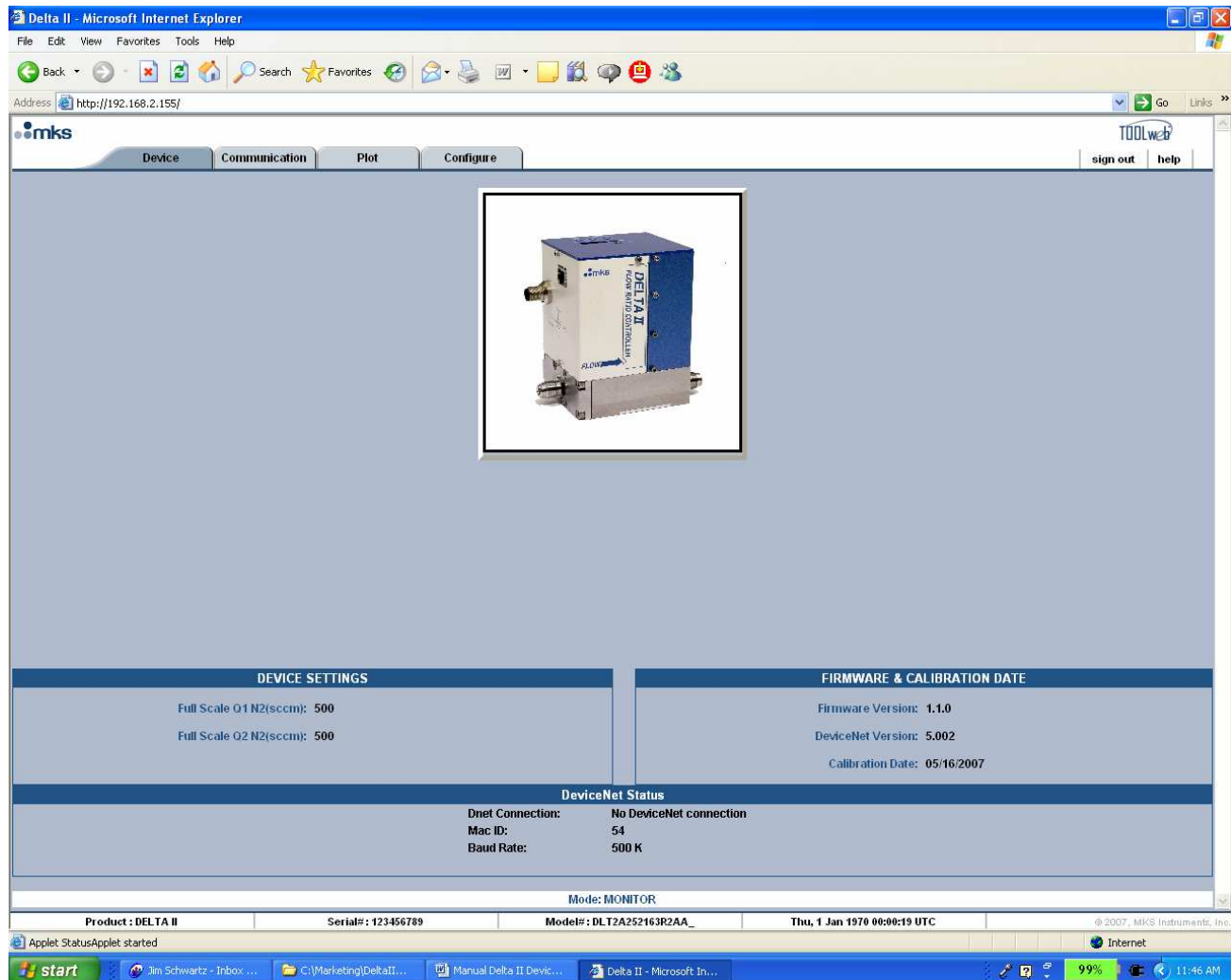


Figure 8: Embedded GUI, Delta II Ratio Controller Device Page in Monitor Mode

Communication Page – Monitor Mode

On the Communication Page is the following information for the Delta II under TCP/IP Information.

- Ethernet MAC Address
- IP Address
- Subnet Mask
- Default Gateway

These settings are not changeable in either the Monitor or Setup Modes for the device.

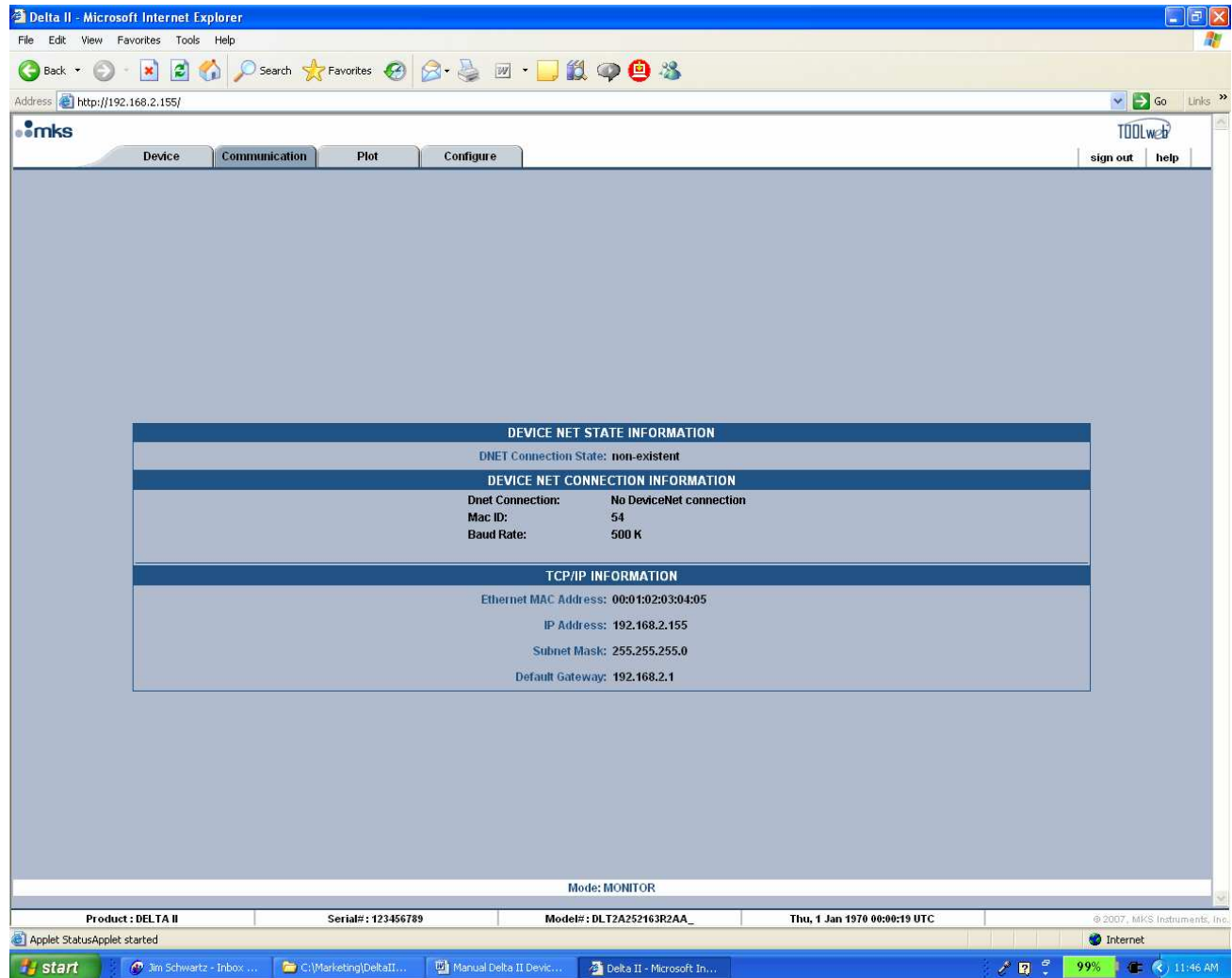


Figure 9: Embedded GUI, Communications Page

Plot Page – Monitor Mode

This page enables you to plot and see real-time performance of the device. In Monitor Mode you are able to select variables to plot, the rate at which to display them, and save the data that has been plotted.



Note

The Plot Page requires a java applet to use the plot program. This applet, which must be installed on the computer trying to view the page, can be downloaded by following the instructions listed on page 23.

Selecting Variables To Plot

On the right-hand side of the Plot page you should see the Variables section. Here you have the ability to select one or more variables to plot.

- *To plot one variable:* click on the variable you wish to plot.
- *To plot two or more variables:* either select the variables one-by-one while holding your keyboard's control "Ctrl" key down or select all the variables by selecting the first variable in the list and then selecting the last variable listed while holding your keyboard's "Shift" key down.

Selecting Rate

Directly below the list of variables is the Rate selection drop down menu. Here you are able to select the sampling rate at which you'd like to plot the variables. Available rates are 1, 2, 5, 10, 50 and 100 Hz.

Starting and Stopping the Plot Program

Below the Rate selection is the Start/Stop button for the plot program. You must click on this button to start the plot program and click on it a second time to stop the program.

Options (Trace Autoscroll, Rescaling Y Axis, Rescaling X Axis, Save to File)

Below the Start/Stop button, you should see the Options section. Here you are able to start/stop the autoscrolling feature, rescale the Y-Axis, rescale the X-Axis or save the plot data to a file. These options are described in detail below:

- Trace Autoscroll → Unchecking this checkbox will stop the plot program from scrolling in the X-direction. This option only has an effect while the program is running. Rechecking this checkbox will enable the plot program to resume scrolling. The X-Axis scroll bar may be used for manual scrolling.
- Rescaling Y-Axis → Next to where it says "Y scale:" enter in the scale (Min) to (Max), and then click on the "Rescale Y-Axis" button. This option only has an effect if the "Trace Autoscroll" checkbox is unchecked or the plot program is stopped. Otherwise the Y-Axis will automatically scale itself to fit all variables being plotted.
- Rescaling X-Axis → Entering a value next to where it says "X scale (seconds):" adjusts the number of seconds spanned across the X-Axis, e.g. entering a value of ten seconds sets the X-Axis so that it will show ten second segments at a time. To use this feature, enter the value and then click on the "Rescale X-Axis" button. This option can be used at anytime.
- Save to File → The "Save to file" option can be used at anytime once you've started the plot program. The data stored consists of the data collected from the time the Start button was pressed to the time the "Save to file" button is pressed. The file will be saved in a (.csv) format which can be later imported into a spreadsheet or other program for further analysis.
- Control Settings: See the explanation provided for these settings in the description provided for the Configure Page, Monitor Mode which follows.

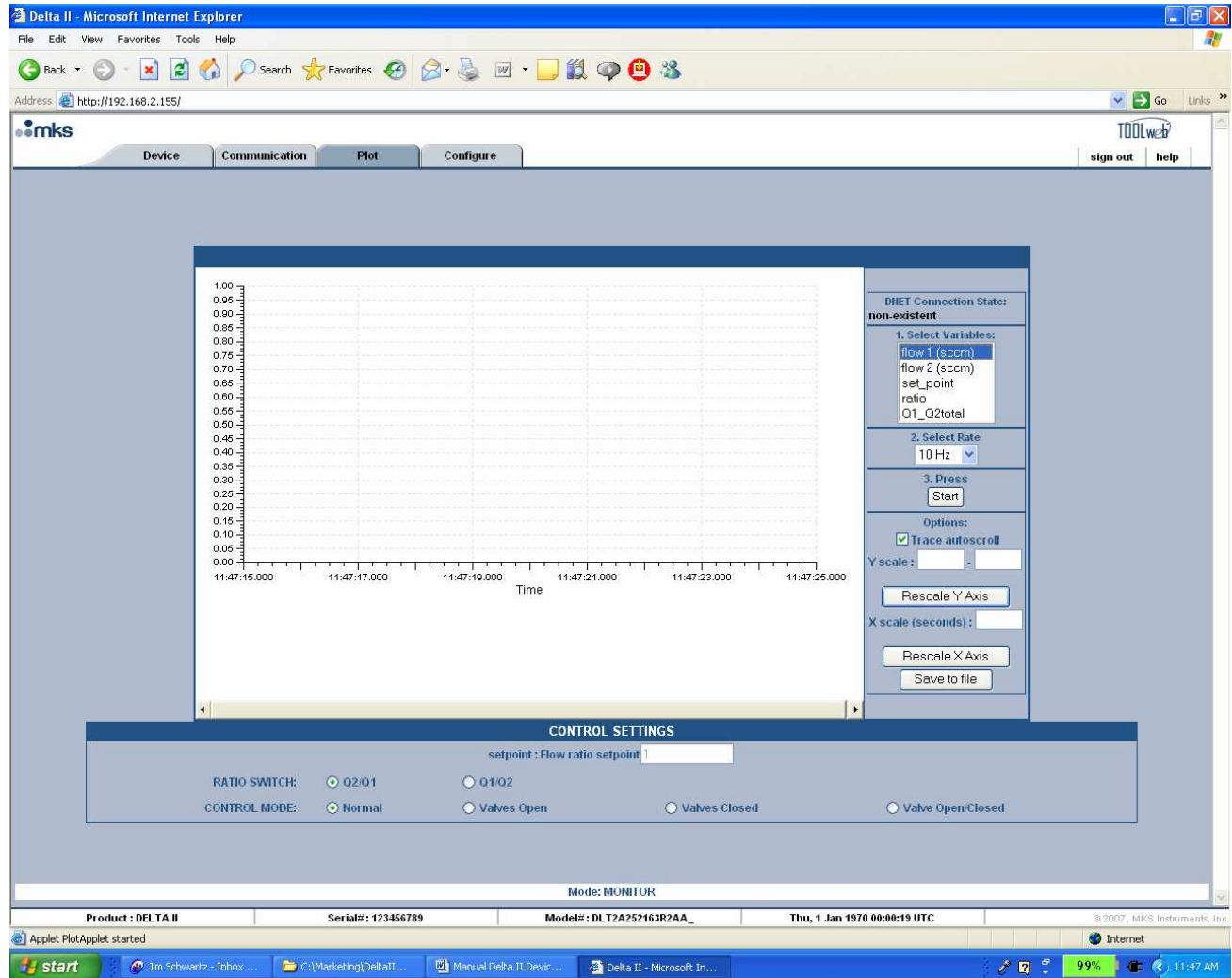


Figure 10: Embedded GUI, Plot Page in Monitor Mode

Configure Page – Monitor Mode

Configuration

Password: On this page you are able to enter the password to change into “Setup Mode.” The Factory-shipped password is “config” (without the “”). Figure 11, below, shows a screen capture of the Configuration Page in “Monitor Mode.”

Control Settings: The settings displayed for the following parameters are provided here for information only. **They may not be changed in the Monitor Mode.**

Setpoint: This is the current setpoint the device is acting on. The device is receiving this setpoint from the host via the Devicenet connection and can not be changed while in the Monitor Mode.

Ratio Switch: This shows which channel is currently considered the control channel as it is listed in the numerator of the active (green button) display.

Control Mode: This displays the current input as to whether the device is controlling to Normal (setpoint provided by the tool), Valves Open, Valves Closed or Valves Open/Closed. For Valves

Open/ Closed, Open refers to the numerator channel listed in the active (green button) displayed while the Closed refers to the other channel.

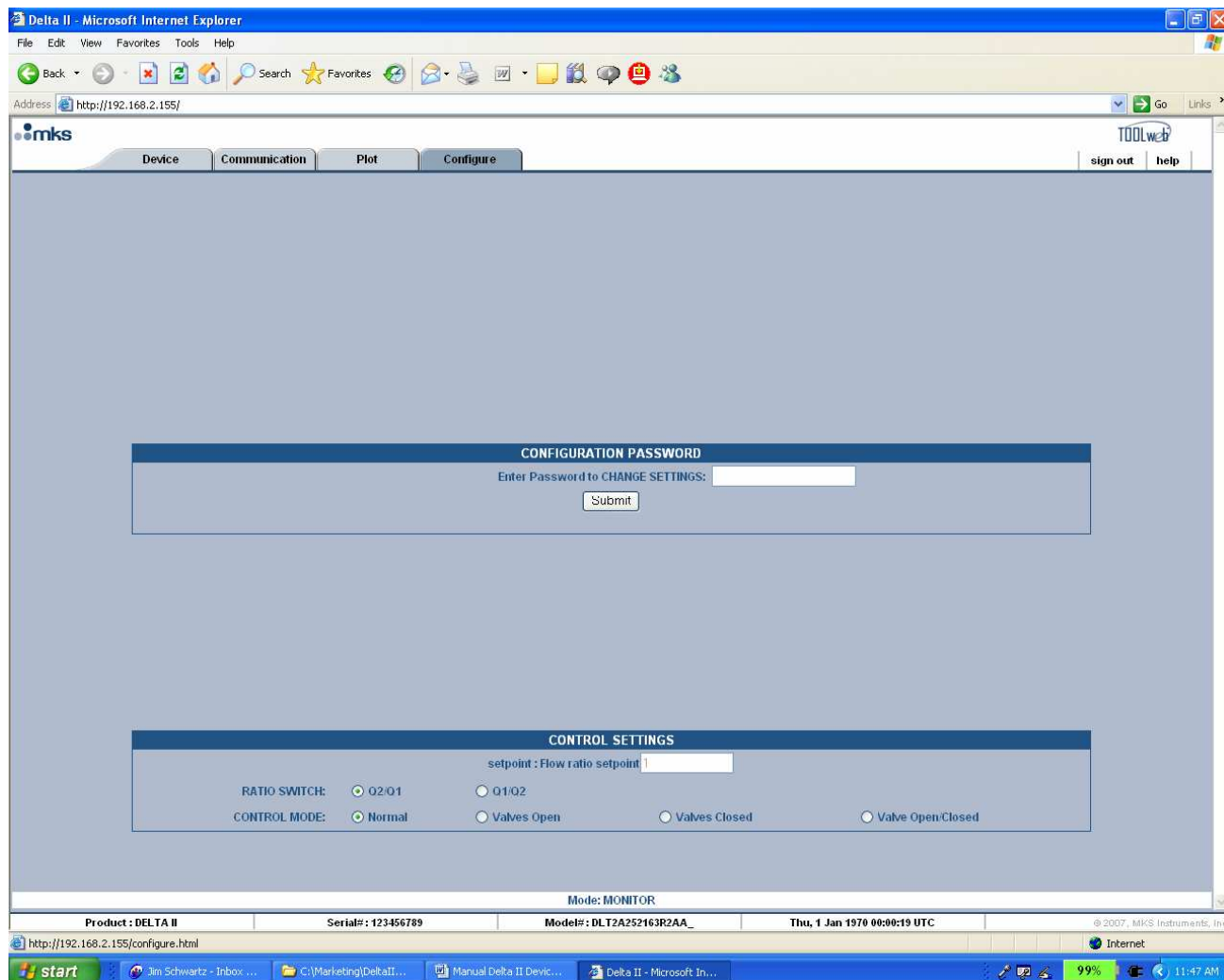


Figure 11: Embedded GUI, Configure Page in Monitor Mode

Setup Mode

In “Setup Mode” the user is able to configure the Delta II Ratio Controller, i.e. zero the device, change the IP address, configure the control parameters, etc. To enter this mode, while in “Monitor Mode”, go to the “Configuration Page” and enter the Factory-shipped password “config” (without “”). Once you press the “Change Settings” button you will be directed to the “Configuration Page” in “Setup Mode.” You should now see a green banner that lines the top of the page that says, “The device is now in SETUP mode.”

Each of the pages, i.e. tabs, in “Setup Mode” are listed and described below in detail starting with the Device Page. Please note that this section will only describe in detail the features that are different from those in “Monitor Mode.” For a complete understanding of each page, also read the *Monitor Mode* section starting on page 35.

A full description of the Configure Page is found later in this Section.

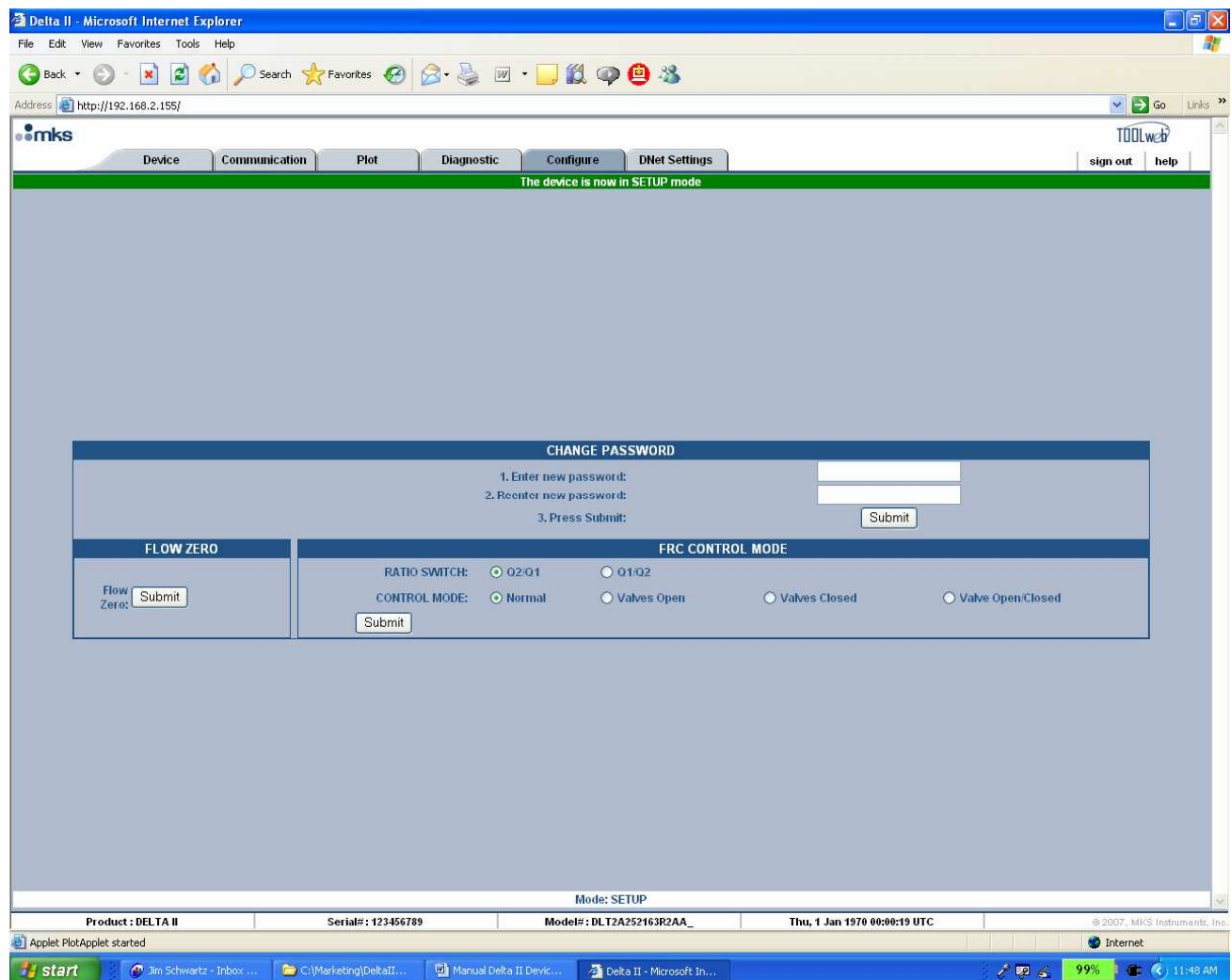


Figure 12: Setup Mode – Configure Page

Device Page – Setup Mode

In the “Setup Mode”, in addition to the Device Settings and Firmware & Calibration Date the Display Mode is available. For Device Settings and Firmware & Calibration Date see the Monitor Mode – Device Page section of this manual.

Display Mode: The display mode refers to the information normally displayed on the scrolling 4 digit LED on the top of the Delta II enclosure after the 60 second power cycle. To select what is shown via the display click on the drop down arrow, select (click on) the item to display and then click on the submit key. The following items are available to be displayed. Only one may be selected at any time.

IP - IP Address

Q1 - Channel flow rate in Nitrogen equivalent sccm

Q2 - Channel flow rate in Nitrogen equivalent sccm

RATIO - Current setpoint

TEMP - Controlling channel temperature (degrees Centigrade)

TEST - Causes all LED pixels to illuminate to check their viability

MKS - MKS banner line scroll

Default - Scrolling display same as on startup

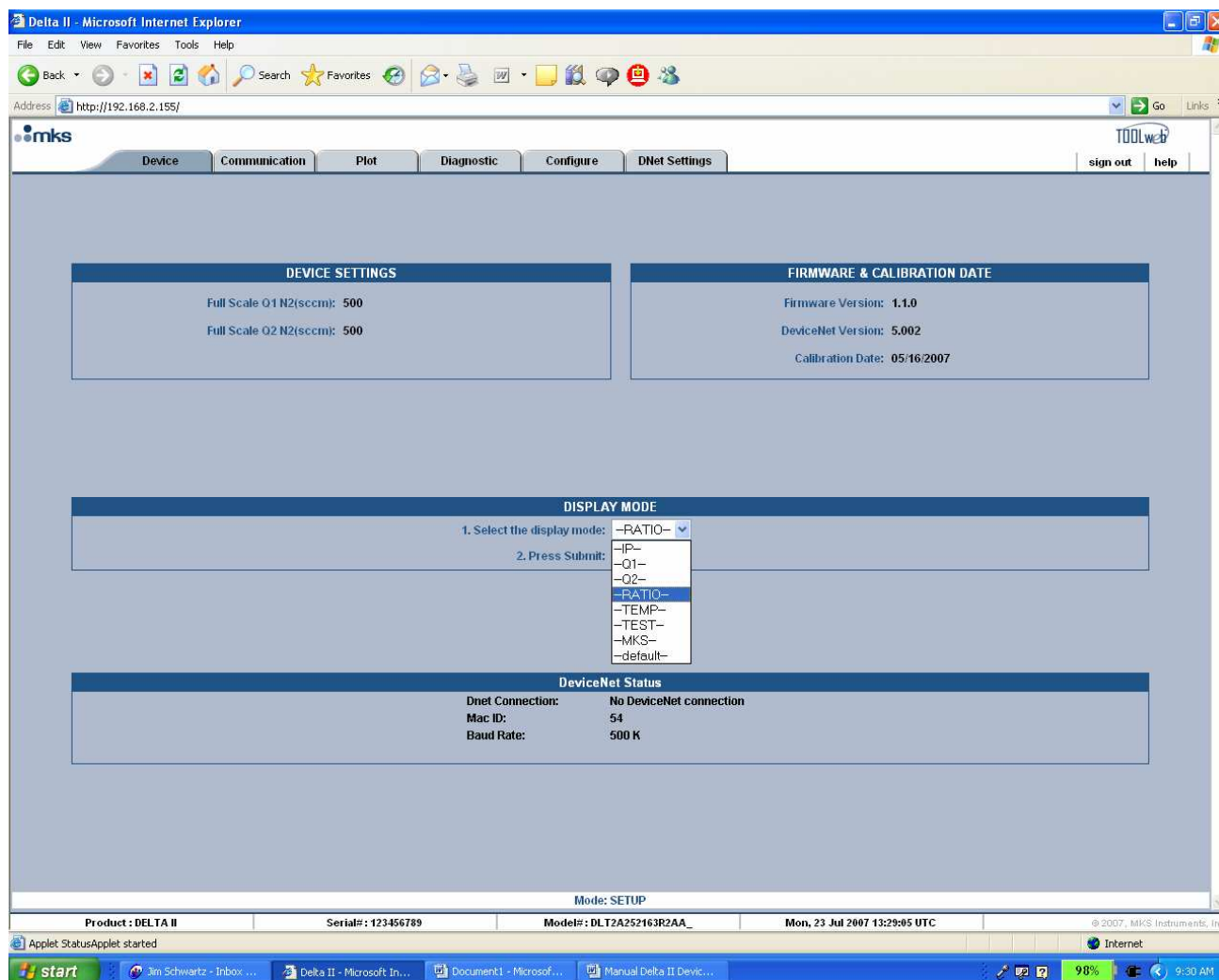


Figure 13: Embedded GUI, Device Page in Setup Mode

Communication Page – Setup Mode

On the Communication Page is the following information under TCP/IP Information.

Ethernet MAC Address

IP Address

Subnet Mask

Default Gateway

These settings are not changeable in either the Monitor or Setup Modes for the device.

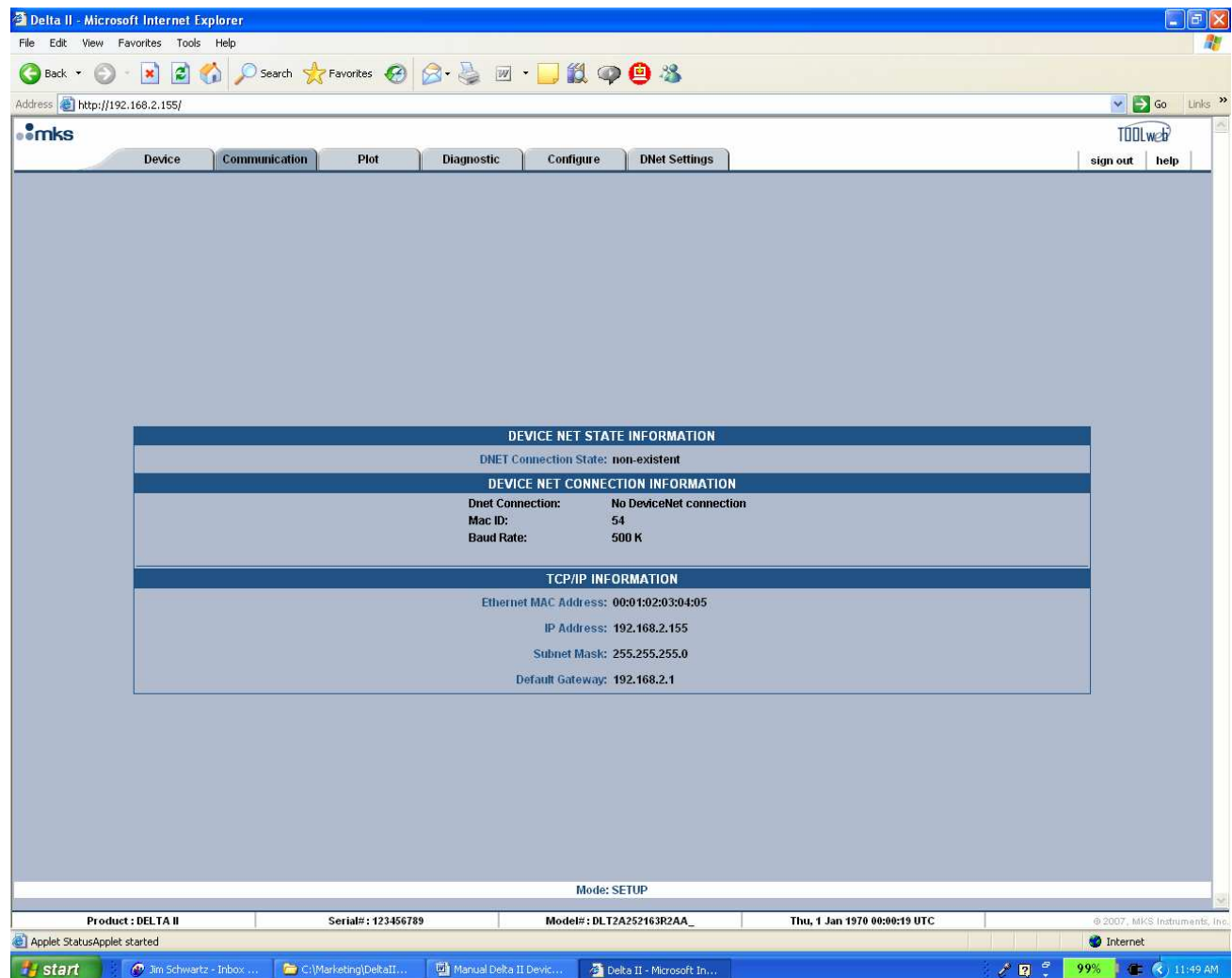


Figure 14: Embedded GUI, Communications Page in Setup Mode

Plot Page – Setup Mode

In “Setup Mode” this page enables the user to either plot performance while the device is under Devicenet (tool or host) control or while the device is under (UI) control.



Note When there is an active Devicenet connection to the Delta II Flow Ratio Controller, the Ethernet UI may not be used to change any device parameters.

Figure 15, below, is a screen capture of the Plot Page in “Setup Mode.” The setpoint section is located above the “Select Variables” section in the top right-hand. This page enables you to send a setpoint to the Delta II Flow Ratio Controller through Ethernet (see note above) when the device is powered but disconnected from the Devicenet network. The setpoint value will be black when it may be given via the UI and “ghosted” when the setpoint is from the active network.. A setpoint can be given by highlighting the Setpoint box, typing in a valid setpoint and submitting it.

Valid ratio setpoints are: $\alpha = 1$ to 20

Other than setting the setpoint in Digital Mode, the user may always plot the device response by highlighting those parameters of interest in the Select Variables box. Plotting parameters and re-ranging the axis is as described in the Monitor Mode – Plot Page section of this manual.

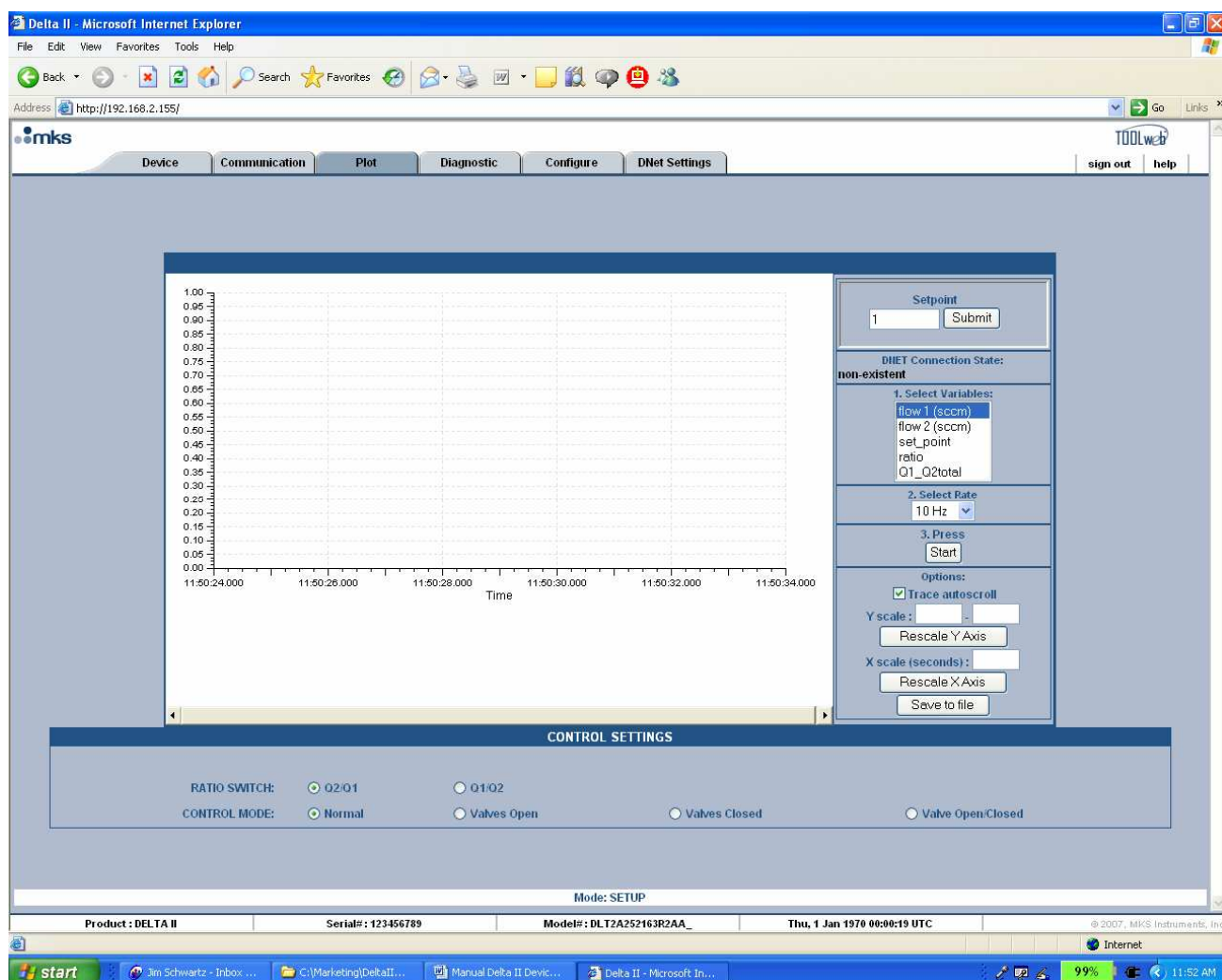


Figure 15: Embedded GUI, Plot Page in Setup Mode

Diagnostics Page – Setup Mode

In the “Device Diagnostics” section the user is given the ability to run basic diagnostic tests on the flow and temperature. These tests can be run individually or all at once.

Device Diagnostics

The tests listed in this section are designed to insure that there are no major electrical problems and that the sensors are working properly. These tests are described in detail below: the “1” and “2” refer to the two flow channels.

- Raw Flow → This basic test is a simple diagnostic check for the flow circuit to verify that the sensor and the electronics are working properly. To run this test, select the checkbox next to where it says “Raw Flow:” and then press the “Run the test” button. When the test finishes you will either see the word “Pass” or “Fail” next to the checkbox. If the test passes, then the flow circuit is good. If it fails, then there may be a problem. See Cycle Power below.
- Temperature → This basic test is a simple diagnostic check for the temperature circuit to verify that the sensor and the electronics are working properly. To run this test, select the checkbox next to where it says “Temperature:” and then press the “Run the test” button. When the test finishes you will either see the word “Pass” or “Fail” next to the checkbox. If the test passes, then the temperature circuit is good. If it fails, then there may be a problem. See Cycle Power below.

Figure 16, below, shows a screen capture of the Diagnostics Page.

Cycle Power

After testing or to reboot the Delta II pressing the “Cycle Power” button will cause the device to go through a power cycle and reset the device. If the device failed either the Raw Flow or Temperature tests above, it is advised that the power be cycled and the test rerun. After pressing “Cycle Power”, the UI may need to be relaunched as described previously. Should the device fail again, contact your MKS Service Center.

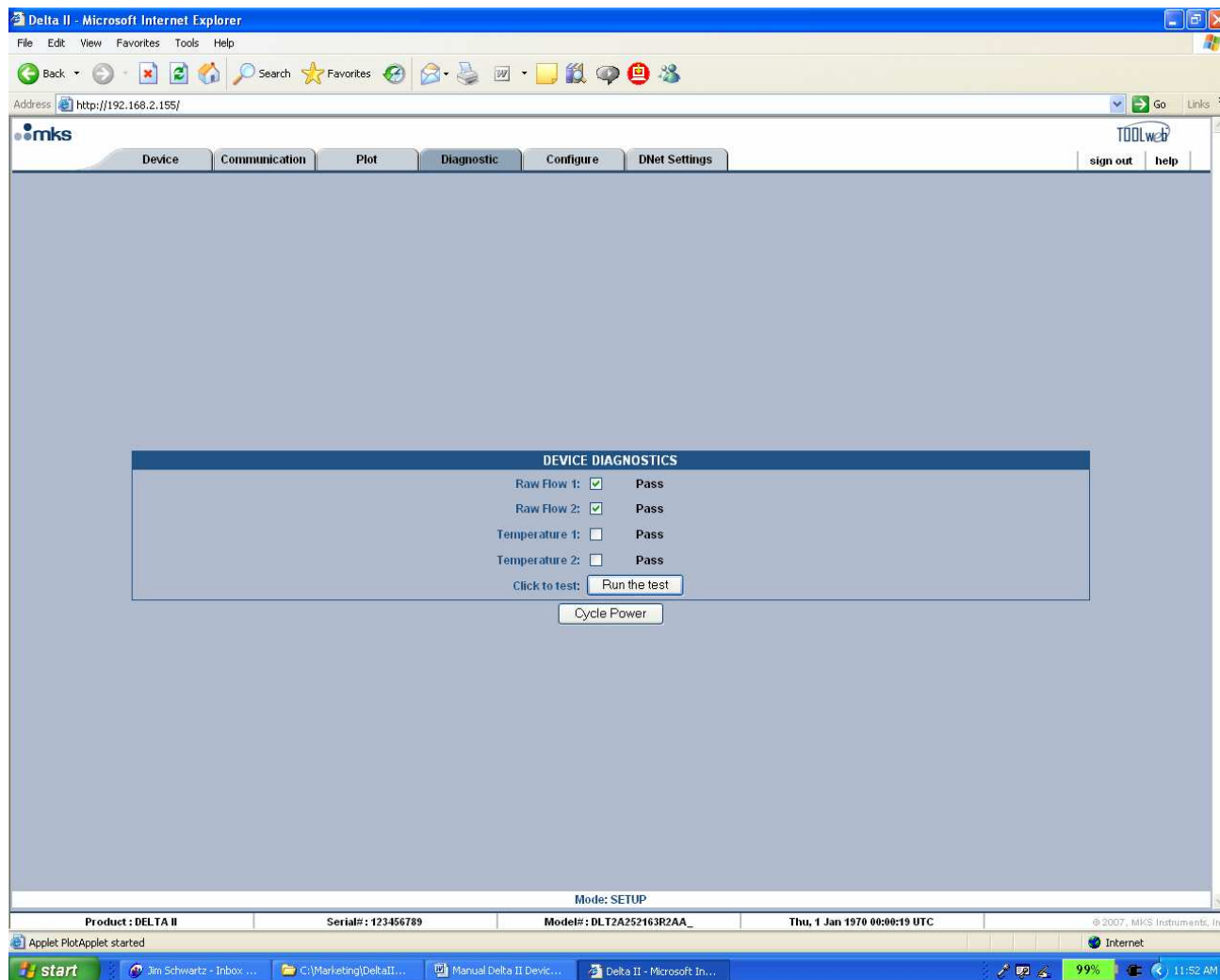


Figure 16: Embedded GUI, Diagnostic Page in Setup Mode

Configuration Page – Setup Mode

As was noted earlier, the Configuration Page is where you are initially directed once you’ve entered “Setup Mode.” Here the Delta II Ratio Controller can be zeroed, the “Setup Mode” password can be changed, the Ethernet settings changed, firmware can be updated and device control mode be changed.



Caution Zeroing the flow incorrectly can cause system failure. Make sure that the Zero Adjustment procedure on page 83 is followed properly.

Changing the Setup Mode Password

To change the “Setup Mode” password from the default password “config” follow the steps below:

1. Under Change Setup Password, enter the new password in the box to the right of line 1.
2. Enter the new password again, this time in the box to the right of line 2.
3. Click on the “Submit” key to enter the new password.

To confirm the new password has been accepted, click on “Sign Out” in the right corner of the page. This will put the device back into the Monitor Mode. Go to the Configure screen and enter the new password. The

device should now change into the Setup Mode. If the device does not go to the Setup Mode, then enter the “old” password to get back to the Setup Mode.

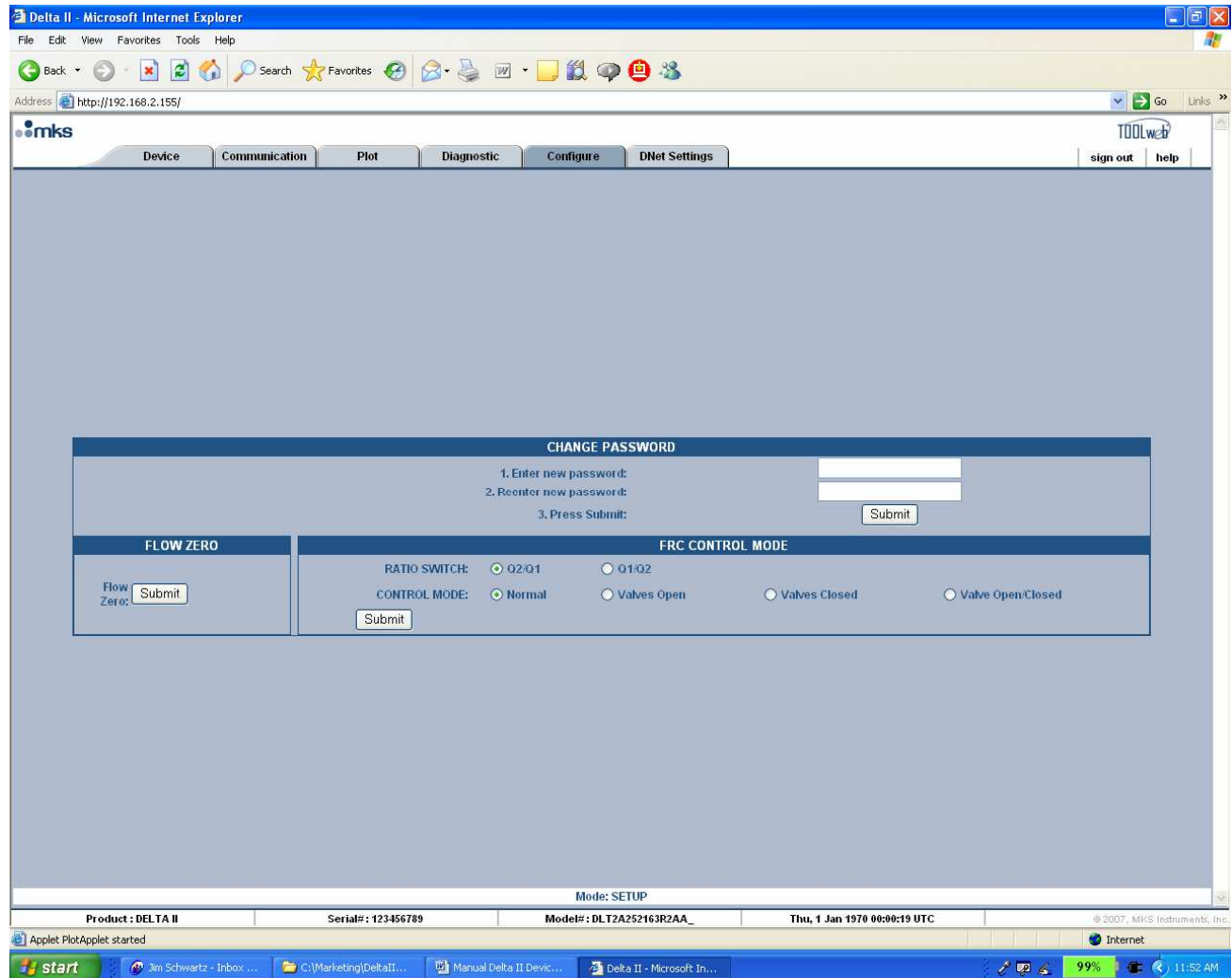


Figure 17: Embedded GUI, Configure Page in Setup Mode

Zeroing the Delta II Flow Ratio Controller

For the Delta II Ratio Controller there is a “Flow Zero” option. Before zeroing the value make sure the Zero Adjustment procedure (page 83) has been followed correctly. To zero the flow, press the “Flow Zero” button. A green banner across the top of the page will tell you if the device received the command and is processing. Zeroing typically will take (10) seconds. The Delta II Ratio Controller will be unresponsive during this time.



Caution

Zeroing the flow incorrectly can cause system failure.. Make sure that the “Zero Adjustment” procedure on page 83 is followed properly.

Changing the Control Mode Delta II Ratio Controller

Ratio Switch: The Ratio Switch is used to select the definition, either Q_2/Q_1 or Q_1/Q_2 , of the ratio setpoint such that the ratio setpoint is between 1 and 20.

Control Mode: The Control Mode is used to effect direct control over the valves in the Delta II. Normal refers to when control is given to the setpoint given to the device as displayed on the Plot Page of the device. Valves Open will cause both valves to immediately go to a full open condition. Valves Closed will cause both valves to immediately go to a full closed position. Valve Open/Closed when selected will cause the valve designated by the numerator in the Ratio Switch selection to go open while the valve designated in the denominator of the Ratio selection will go closed.

DNet Settings Page – Setup Mode

The DNet Settings Page displays (in “ghosted” characters) the current Flow 1, Flow 2, Temperature and Other Parameters. This information is displayed for information only and is not settable through the Setup Mode. Parameters that may be changed by the user must be done through the Devicenet connection.

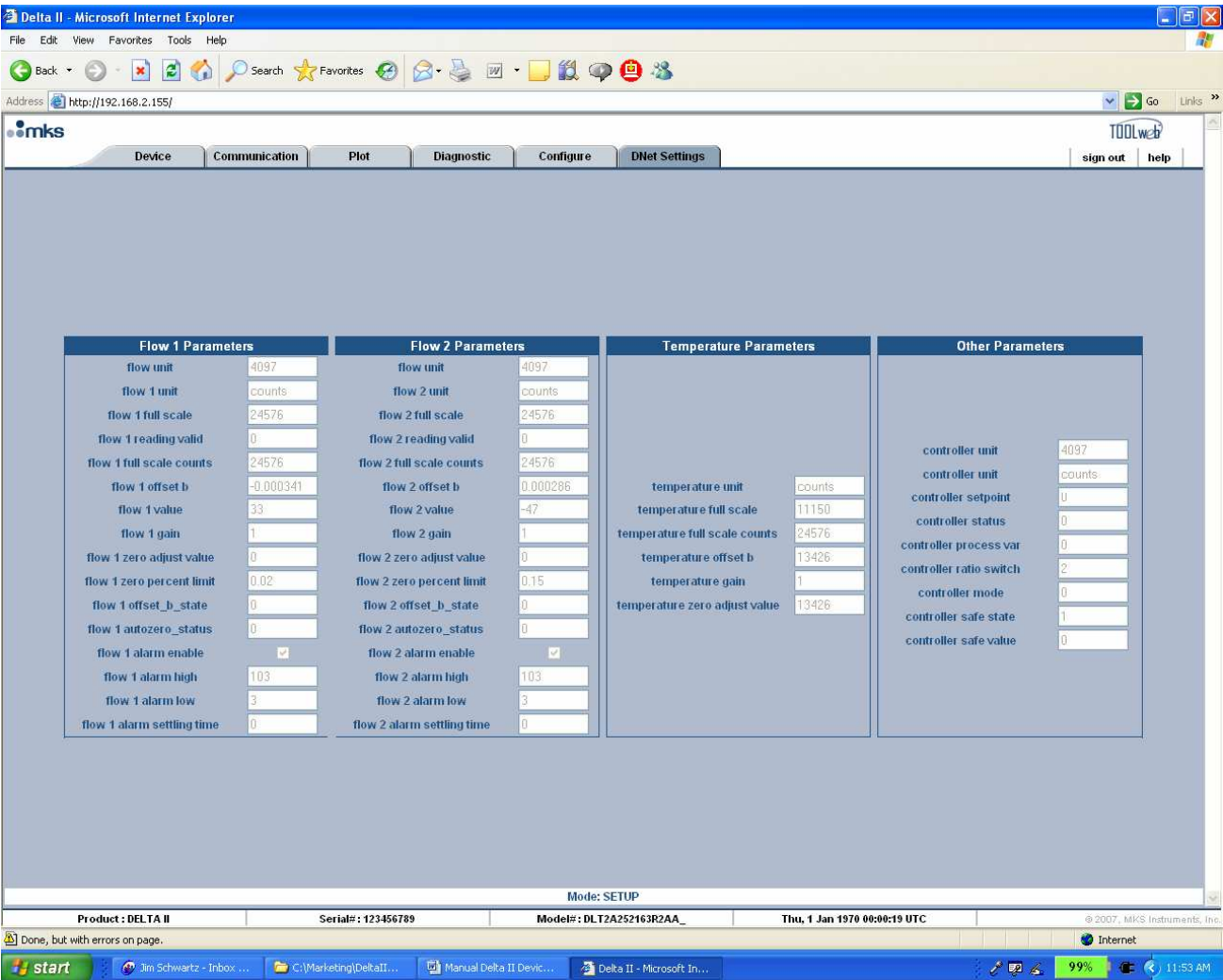


Figure 18: Embedded GUI, DNet Settings Page in Setup Mode

Chapter Five: Operation

General Information

The MKS Delta II Flow Ratio Controller is used in applications that require the delivery of a gas or combination of gasses to multiple locations at controlled ratios. For example, the same amount of a gas mixture can be delivered to more than one wafer at a time. Also, the uniformity on a wafer may be improved by controlling the flow to different areas of the wafer.

The total flow is controlled upstream of the MKS Delta II Flow Ratio Controller (by other system components) which is then divided by the MKS Delta II Flow Ratio Controller in response to programmed ratio setpoints (Alpha) as the process requires. The ratio setpoint, α , is given to the device using the Devicenet communications protocol as defined in Chapter 6, Devicenet Protocol.

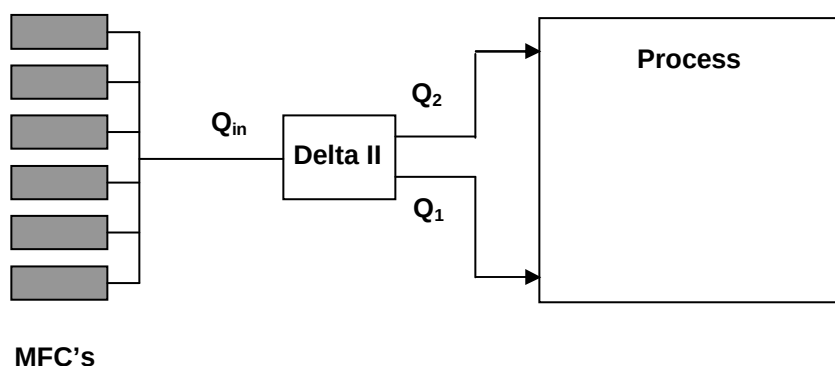


Figure 19: FRC Application Block Diagram

In the case of the MKS Delta II Flow Ratio Controller, Q is the mass flow into the device, Q_1 is the flow in channel 1 and Q_2 is the flow in channel 2. Normally, Alpha, α , is equal to Q_2/Q_1 .

$$\alpha = Q_2/Q_1$$

When the Ratio Switch command is active, Alpha, α , is equal to Q_1/Q_2 .

$$\alpha = Q_1/Q_2$$

The MKS Delta II Flow Ratio Controller achieves alphas from 1 to 20 in both cases for units with identical full-scale channels. The MKS Delta II Flow Ratio Controller is gas independent. The system is fed a single alpha command and the error is calculated by subtracting the measured alpha from the alpha set point.

The MKS Delta II Flow Ratio Controller controls alpha using two thermal based flow sensors typically used in MFC and MFM products. One is installed on each flow channel. Also on each flow channel is a controllable valve. From a control stand point the valve behaves as a variable orifice. Figure 20 depicts the relationship between the flow in each channel when ratio, $\alpha = Q_2/Q_1$.

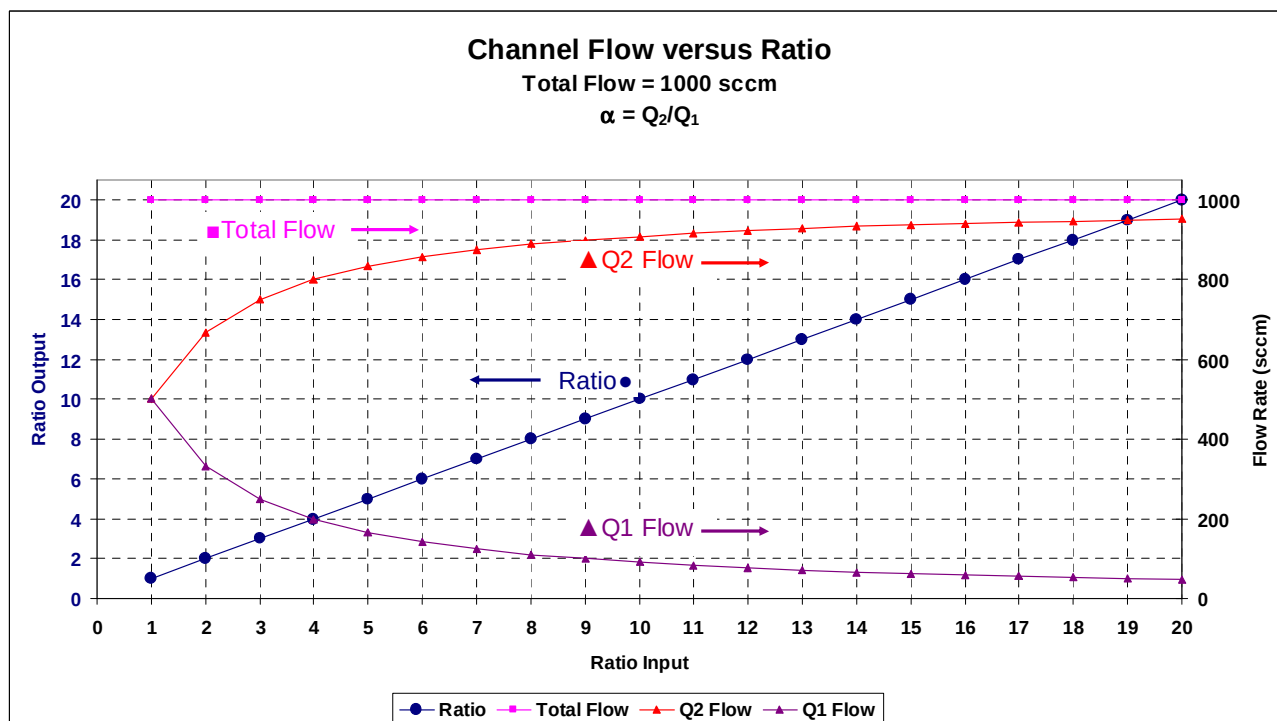


Figure 20: Channel Flow as a Function of Ratio Input at 1000 sccm Input Flow

Control System

The control system in the MKS Delta II Flow Ratio Controller is a multi-input multi-output system. Depending on alpha, the controller commands one valve to open fully. The ratio of the measured flows is used to derive the command for control of the second valve. This basic control concept of controlling the flow ratio, by definition, is gas independent. The errors in the flow measurements due to differing gasses are the same for both sensors and thus the ratio of the orifices is correct. The 'fixed' and 'control' valve functions are inverted when the Ratio Switch command is active (available on units with identical full-scale flow channels only) which allows 1:20 and 20:1 ratio control.

Control I/O

The Delta II Flow Ratio Controller device operates as a slave on a Devicenet[®] network. The unit supports Explicit Messages and Polled I/O Messages of the predefined master/slave connection set. It does not support the Explicit Unconnected Message Manager (UCMM). See Chapter 6 for details of the communication protocol.

Warm Up Time

After installation and power up, allow the MKS Delta II Flow Ratio Controller to warm up for a minimum of 60 minutes.

Priority of Commands

The MKS Delta II Flow Ratio Controller has an established hierarchy that it uses to determine which operating mode takes precedence. The operating modes are listed according to the order of priority (from highest to lowest):

- Valves Closed / Valves Open
- Closed-Loop Control Operating Mode (Normal Ratio control)

For example, with a setpoint established into the unit, the valve open or close command may be used to immediately open or close the proportional valves in the unit.

Closed-Loop Control Operating Mode

When the MKS Delta II Flow Ratio Controller operates in closed-loop control mode, it compares the ratio signal from the Flow Ratio Readback to the Flow Ratio Setpoint. The MKS Delta II Flow Ratio Controller positions its valves to achieve or maintain the setpoint flow ratio in the system. Closed-loop control has the lowest priority. The MKS Delta II Flow Ratio Controller defaults to closed-loop control in the absence of any higher priority commands.

The Gas Correction Factor (GCF)

A Gas Correction Factor (GCF) normally used with mass flow sensors, is used to indicate the ratio of flow rates of different gases which will produce the same output voltage from the mass flow sensor. The GCF is a function of specific heat, density, and the molecular structure of the gases. Nitrogen is used as the baseline gas (GCF = 1) since flow controllers are usually calibrated with nitrogen.

Since the MKS Delta II Flow Ratio Controller uses two matched flow sensors and both sensors are subjected to the same pressures, they both respond in the same manner to the gas flowing through them. The gas flowing through both sensors is the same. The difference between the sensors is used to derive the ratio flow between them and as a consequence, the absolute flow is not required to get an accurate ratio value. The accuracy of the ratio depends on the match between the sensors.

The GCF is, however, useful in the proper selection of the FRC with respect to the Full-scale operating region.

Devicenet Controls and Indicators

The top panel, shown in Figure 4, page 18, of the MKS Delta II Flow Ratio Controller contains several Devicenet controls and indicators.

The MKS Delta II Flow Ratio Controller has two standard bi-color (green/red) Devicenet status LEDs, (Module Status LED and Network Status LED) located on top of the instrument, as shown in Figure 4, page 18. The power-up sequence of these LEDs conforms to the requirements in the ODVA Devicenet Specification, Volume I [1].

In addition to the LEDs, the MKS Delta II Flow Ratio Controller has three rotary switches used to control the unit's communication state (a Baud Rate switch and two MAC ID switches).

Module Status LED

The Module Status LEDs indicate the status of each of the individual network nodes, as defined in Table 5.

Table 5: Module Status LED Indicators

LED State	Meaning
Solid Green	System Functioning Normally
Flashing Green	Wink ON
Flashing Red	Major Recoverable Fault
Solid Red	Major Unrecoverable Fault
Dark	Power is OFF
Flashing Red/Green	Self testing



Note

A major unrecoverable fault prevents operation because the device cannot communicate on the network. Contact MKS Instruments, Inc. for assistance.

Network Status LED

The Network Status LED indicates the status of the communications link, as defined in the table below. If no problems are detected, the Network Status LED illuminates a green. A solid red or dark Network Status LED indicates a fault condition.

Table 6: Network Status LED Indicators

LED State	Meaning
Solid Green	Communications link is OK. The device is online and connections are established.
Flashing Green	The device is online but no connections are established. The device has passed the Dup_MAC_ID test and is online, but has no established connections to other nodes.
Solid Red	Critical Link Failure. The device has detected an error that prevents network communication (Duplicate MAC_ID or bus-off).
Dark	Not powered / Not online The device has not completed the Dup_MAC_ID test, or the device is not powered; check the module status LED.

Baud Rate and MAC ID Switches

The baud rate and MAC ID (node address) for your device can be set through software commands using standard Devicenet protocol over the network, or manually using the rotary switches located on the top panel of the device. The baud rate and MAC ID switches allow you to easily configure units without an operational network, or to network multiple units quickly.

The baud rate and MAC ID rotary switches support an assigned *network* position, labeled on the device as “PGM” to indicate software operation.

If the rotary switch is in the network (PGM) position at power-up, the baud rate or address is read from the non-volatile memory. Any changes to the values must be made over the network; any changes in the rotary switch positions after power-up are ignored.

If the rotary switch is *not* in the network (PGM) position at power-up, the baud rate or address is read directly from the switches.

Baud Rate Switch

The 4-position rotary switch, shown in Figure 4, page 18, is used to select the Devicenet baud rate. The choices are: PGM (the baud rate is read from the non-volatile memory), 125, 250, and 500 Kb. The switch positions are numbered in a clockwise direction, to correspond to the increasing baud rate values.

MAC ID (Node Address) Switches

Two 10-position rotary switches, shown in Figure 4, page 18, are used to set the MAC ID (node address).

The MAC ID is an integer identification value assigned to each node on the Devicenet network.

The valid MAC ID switch positions are 0 to 63 although 0 is usually reserved for the Devicenet master. Use the switch on the left to set the most significant digit (MSD), that is, the factor of ten (10, 20, 30...60). Use the switch on the right to set the least significant digit (LSD), that is, the increments of one (1, 2, 3...9). The switch positions are numbered in a clockwise direction, to correspond to the increasing address values.



Note

Setting the switches to a value that is greater than 63 is the same as setting the rotary switch to the “PGM” position (the baud rate is read from the non-volatile memory).



Note

The MAC ID switch on the top of the device must be set to the network (PGM) position at power up in order for changes to be made over the network. Any changes in the rotary switch positions after power up are ignored.

Power Up

At power-up, the device performs internal diagnostic checks of the EEPROM and RAM and checks on its communications link. The results of these checks are indicated by the color (green or red) and condition (solid or flashing) of the status LEDs on top of the instrument. The following LED sequence occurs when the DPC is powered up (times are approximate):

1. The Module Status LED flashes first GREEN for ¼ second, then RED for ¼ second, then turns OFF.
2. The Network Status LED flashes first GREEN for ¼ second, then RED for ¼ second, then turns OFF.
3. The Module Status LED flashes from GREEN to RED while the device is initializing. The Network Status LED remains OFF.
4. The Module Status LED illuminates solid GREEN when initialization is complete.
5. When the device establishes communication with other devices on the network, the Network Status LED illuminates GREEN.



Note

If the power up LED sequence does not function properly, contact MKS Instruments, Inc. for assistance.

See Figure 4, page 18, for the location of the Module Status LED and Network Status LED. For more information on the operation of the LEDs, see page 52.

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Chapter Six: Devicenet Protocol

Use this manual with the ODVA Devicenet Specification Volume I and Volume II [1, 2], and the SEMI Standards Common and Specific Device Models [3, 4]. Refer to those documents for a complete functional description of the MKS Delta II Flow Ratio Controller.

This chapter defines the application specific objects used by MKS Delta II Flow Ratio Controller. It also defines how system requirements are mapped to specific objects and attributes in the Devicenet protocol. Tables defining the attributes and supported services for each object model are provided. The objects, attributes, and services described in this manual comply with the definition of an interoperable device on a semiconductor equipment sensor/actuator network proposed by SEMI [3, 4].

The Delta II Flow Ratio Controller device operates as a slave on the Devicenet[®] network. The unit supports Explicit Messages and Polled I/O Messages of the predefined master/slave connection set. It does not support the Explicit Unconnected Message Manager (UCMM).

Object Models

The object models supported by the MKS Delta II Flow Ratio Controller are listed in the table below. The attributes and services associated with each object model are defined in this chapter.

Descriptions are supplied throughout this document for attributes and services that are either vendor specific, or require additional specifications beyond the Devicenet and SEMI specifications [1,2,3,4].

Table 7: Object Modules Present in the Delta II Flow Ratio Controller

Object	Class Code Hex (dec)	Instances dec	Function
Identity	0x01 (1)	1	Provides identification and general information about the device
Message Router	0x02 (2)	1	Routes incoming messages to the appropriate Devicenet object
Devicenet	0x03 (3)	1	Provides the configuration and status of the physical connection to Devicenet
Assembly	0x04 (4)	11 Instances of Static Input: 2 Instances of Static Output:	Binds attributes of multiple Devicenet objects allowing data to be sent or received over a single connection.
Connection	0x05 (5)	1,2	Manages the internal resources associated with the I/O and Explicit Messaging connections
S-Device Supervisor	0x30 (48)	1	Centralizes SEMI device state definitions, status, and behavior.
S-Analog Sensor	0x31 (49)	1,2	Models the acquisition of a reading from the mass flow sensor.
S-Single Stage Controller	0x33 (51)	1	Models the closed-loop control system of the mass flow device.

Devicenet Message Types

As a group 2 slave device the Flow Ratio Controller supports the following message types.

Table 8: Devicenet Message Types

CAN IDENTIFIER	GROUP 2 Message Type
10xxxxxx111	Duplicate MACID Check Message
10xxxxxx110	Unconnected Explicit Request Message
10xxxxxx101	Master I/O Poll Command Message
10xxxxxx100	Master Explicit Request Message
xxxxxx = Node Address	

Identity Object: Class Code 01 (0x01)

The Identity Object is required on all devices and provides identification of and general information about the device.

Class Attributes

Table 9: Identity Object Class Attributes

Attribute	Access	Name	Type	Value
1	Get	Revision	UINT	1

Instance 1 Attributes

Table 10: Identity Object Instance 1 Attributes

Attribute	Access	NV	Name	Type	Value
1	Get	NV	Vendor	UINT	36 = MKS Inst.
2	Get	NV	Device Type	UINT	100 = Vendor Specific
3	Get	NV	Device Code	UINT	6
4	Get	NV	Revision	STRUCT OF	See Below
			Major Revision	USINT	
			Minor Revision	USINT	
5	Get	V	Status	UINT	See Below
6	Get	NV	Serial Number	UDINT	See Below
7	Get	NV	Product Name	SHORT_STRING	Delta II
8	Get	V	State	USINT	See Below
*** This attribute resides on the Controller's memory.					

Common Services

Table 11: Identity Object Common Services

Service Code	Class	Instance	Service Name
05 (0x05)	No	Yes	Reset
14 (0x0E)	Yes	Yes	Get_Attribute_Single

Revision (Attribute 4)

MKS maintains strict revision control. The major revision number will increment as functional enhancements are implemented. The minor revision will increment if minor changes are incorporated.

Device Status (Attribute 5)

Table 12: Identity Object Device Status

Bit	Meaning	Status
0	owned	0=not owned 1=owned (default)
1	reserved	0
2	configured	0
3	reserved	0
4-7	vendor specific	0
8	minor cfg. fault	0=no fault 1=minor fault
9	minor dev. fault	0=no fault 1=minor device fault
10	major cfg. fault	0=no fault 1=major cfg. fault
11	major dev. fault	0=no fault 1=major device fault
12-15	reserved	0

Serial Number (Attribute 6)

The serial number is encoded in the product during the manufacturing cycle and is guaranteed to be unique across all product lines produced by MKS.

State (Attribute 8)

The Device State reflects whether any errors have occurred and the severity. The following states are supported. The only exit from a Major Unrecoverable fault condition is power cycling the device.

Table 13: Identity Object State

State	Interpretation	Causes
0	Non-existent	
1	Device Self Testing	
2	Standby	
3	Operational	Normal operating mode
4	Major Recoverable fault	See S_Device_Supervisor Alarm Detail
5	Major Unrecoverable fault	See S_Device_Supervisor Alarm Detail

Controller Revision (Attribute 100)

MKS Instruments maintains strict revision control. The major revision number will increment as functional enhancements are implemented to the Controller firmware. The minor revision will increment if minor changes are incorporated to the Controller Firmware.

Controller Serial Number (Attribute 101)

The Serial Number for the Controller target board will be formatted the same as the attribute 6.

Controller Model Number (Attribute 102)

The Model Number for the Controller target board will be a short string with a max length of 12 characters.

Router Object: Class Code 02 (0x02)

The Message Router Object provides a messaging connection point through which a Client may address a service to any object class or instance residing in the physical device.

Class Attributes**Table 14: Router Object Class Attributes**

Attribute	Access	Name	Type	Value
1	Get	Revision	UINT	1

Instance 1 Attributes**Table 15: Router Object Instance 1 Attributes**

Attribute	Access	Name	Type	Value
2	Get	Number of Connections	UINT	2

Common Services

Table 16: Router Object Common Services

Service Code	Class	Instance	Service Name
14 (0x0E)	Yes	Yes	Get_Attribute_Single

Devicenet Object: Class Code 03 (0x03)

Class Attributes

Table 17: Devicenet Object Class Attributes

Attribute	Access	Name	Type	Value
1	Get	Revision	UINT	1

Instance 1 Attributes

Table 18: Devicenet Object Instance 1 Attributes

Attribute	Access	Name	Type	Value
1	Get/Set	MACID	USINT	See Below
2	Get/Set	Baud Rate	USINT	See Below
3	Get/Set	Bus Off Interrupt	BOOL	See Below
4	Get/Set	Bus Off Counter	USINT	See Below
5	Get	Allocation Information	STRUCT of	See Below
		Choice Byte	BYTE	
		Master Node Addr.	USINT	
6	Get	Mac Switch Changed	BOOLEAN	0=no, 1=yes
7	Get	Baud Switch Changed	BOOLEAN	0=no, 1=yes
8	Get	Current Mac Switch	USINT	
9	Get	Current Baud Switch	USINT	

Common Services

Table 19: Devicenet Object Common Services

Service Code	Class	Instance	Service Name
14 (0x0E)	Yes	Yes	Get_Attribute_Single
16 (0x10)	No	Yes	Set_Attribute_Single
75 (0x4B)	No	Yes	Allocate Master/Slave
76 (0x4C)	No	Yes	Release Master/Slave

MAC ID (Attribute 1)

The MACID is set using two BCD rotary switches located on the front panel. Valid MACID addresses are 0 to 63 (0 to 3F Hex). Setting the switch address to a value greater than 63 will disable the switch and allow software setting of the MACID. The software setting defaults to the last hardware setting. The switch is only read during power up.

Baud Rate (Attribute 2)

Settable only if the Baud Rate switch is set to a value greater than 2. Value returned will be switch value if less than 4 or the last value set.

Table 20: Devicenet Object Baud Rate

Switch/Value	Speed
0	125 kbits
1	250 kbits
2	500 kbits
3	Software settable

Buss Off Interrupt (Attribute 3)

Bus Off Interrupt (BOI) determines the action if a Bus Off state is encountered.

Table 21: Devicenet Object Buss Off Interrupt

BOI	Action
0	Hold chip in OFF state (default)
1	If possible reset CAN chip

Buss Off Counter (Attribute 4)

Bus Off Counter will be forced to 0 whenever set regardless of the data value provided.

Allocation Information – Attribute 5**Table 22: Allocation Choice Byte**

Bit	Meaning
0	explicit set to 1 to allocate
1	polled set to 1 to allocate
2	strobed (not supported)
3-7	reserved (always 0)

Assembly Object: Class Code 04 (0x04)

The Assembly Objects bind attributes of multiple objects to allow data to or from each object to be sent or received over a single connection.

Class Attributes

Table 23: Assembly Object Class Attributes

Attribute	Access	Name	Type	Value
1	Get	Revision	UINT	1
2	Get	Max Object Instance	UINT	13

Instance 1 Attributes

Table 24: Assembly Object Instance 1 Attributes

Attribute	Access	Name	Type	Value
3***	Get	Ratio Readback	INT	See S-Single State Controller, Attribute 7
*** This attribute resides on the Controller's memory.				

Instance 2 Attributes

Table 25: Assembly Object Instance 2 Attributes

Attribute	Access	Name	Type	Value
3***	Get	Data	STRUCT of	
		Exception Status	BYTE	See S-Device Supervisor, Attribute 12
		Ratio Switch	USINT	See S-Single Stage Controller, Attribute 108
		Ratio Readback	INT	See S-Single Stage Controller, Attribute 7
*** This attribute resides on the Controller’s memory.				

Instance 3 Attributes

Table 26: Assembly Object Instance 3 Attributes

Attribute	Access	Name	Type	Value
3***	Get	Data	STRUCT of	
		Exception Status	BYTE	See S-Device Supervisor, Attribute 12
		Ratio Readback	USINT	See S-Single Stage Controller, Attribute 7
		Ratio Setpoint	INT	See S-Single Stage Controller, Attribute 6
*** This attribute resides on the Controller's memory.				

Instance 4 Attributes**Table 27: Assembly Object Instance 4 Attributes**

Attribute	Access	Name	Type	Value
3	Get/Set	Data	STRUCT of	
		Ratio Switch	USINT	See S-Single Stage Controller, Attribute 108
		Ratio Setpoint	INT	See S-Single Stage Controller, Attribute 6

Instance 5 Attributes**Table 28: Assembly Object Instance 5 Attributes**

Attribute	Access	Name	Type	Value
3***	Get	Exception Status	BYTE	See S-Device Supervisor, Attribute 12
*** This attribute resides on the Controller's memory.				

Instance 9 Attributes**Table 29: Assembly Object Instance 9 Attributes**

Attribute	Access	Name	Type	Value
3***	Get	Ratio Readback	REAL	See S-Single Stage Controller, Attribute 7
*** This attribute resides on the Controller's memory.				

Instance 10 Attributes**Table 30: Assembly Object Instance 10 Attributes**

Attribute	Access	Name	Type	Value
3***	Get	Data	STRUCT of	
		Exception Status	BYTE	See S-Device Supervisor, Attribute 12
		Ratio Switch	USINT	See S-Single Stage Controller, Attribute 108
		Ratio Readback	REAL	See S-Single Stage Controller, Attribute 7
*** This attribute resides on the Controller’s memory.				

Instance 11 Attributes**Table 31: Assembly Object Instance 11 Attributes**

Attribute	Access	Name	Type	Value
3***	Get	Data	STRUCT of	
		Exception Status	BYTE	See S-Device Supervisor, Attribute 12
		Ratio Readback	REAL	See S-Single Stage Controller, Attribute 7
		Ratio Setpoint	REAL	See S-Single Stage Controller, Attribute 6
*** This attribute resides on the Controller’s memory.				

Instance 12 Attributes**Table 32: Assembly Object Instance 12 Attributes**

Attribute	Access	Name	Type	Value
3	Get	Data	STRUCT of	
		Ratio Switch	USINT	See S-Single Stage Controller, Attribute 108
		Ratio Setpoint	REAL	See S-Single Stage Controller, Attribute 6

Instance 13 Attributes**Table 33: Assembly Object Instance 13 Attributes**

Attribute	Access	Name	Type	Value
3***	Get	Data	STRUCT of	
		Sensor Value 1	REAL	See S-Analog Sensor, Instance 1, Attribute 6
		Sensor Value 2	REAL	See S-Analog Sensor, Instance 2, Attribute 6
*** This attribute resides on the Controller’s memory.				

Instance 14 Attributes

Table 34: Assembly Object Instance 14 Attributes

Attribute	Access	Name	Type	Value
3***	Get	Data	STRUCT of	
		Exception Status	BYTE	See S-Device Supervisor, Attribute 12
		Sensor Value 1	REAL	See S-Analog Sensor, Instance 1, Attribute 6
		Sensor Value 2	REAL	See S-Analog Sensor, Instance 2, Attribute 6
		Ratio Readback	REAL	See S-Single Stage Controller, Attribute 7
*** This attribute resides on the Controller’s memory.				

Common Services

Table 35: Assembly Object Common Services

Service Code	Class	Instance	Service Name
14 (0x0E)	Yes	Yes	Get_Attribute_Single
16 (0x10)	No	Yes	Set_Attribute_Single

Assembly Instance Details

Assembly instance 1, 2, 3, 5, 9, 10, 11, 13, and 14 are used to generate the POLL response packet and consists of the data described in the table below.

- Assembly instance 4 and 12 are used to consume the POLL request packet and consists of the data described in the table below.

Table 36: Assembly Instance Details

Instance	Byte	Data
1	0	Ratio Readback (low byte)
	1	Ratio Readback (high byte)
2	0	Exception Status
	1	Ratio Switch
	2	Ratio Readback (low byte)
	3	Ratio Readback (high byte)
3	0	Exception Status
	1	Ratio Readback (low byte)
	2	Ratio Readback (high byte)
	3	Ratio Setpoint (low byte)
	4	Ratio Setpoint (high byte)

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Table 36: Assembly Instance Details (continued)

Instance	Byte	Data
4	0	Ratio Switch
	1	Ratio Setpoint (low byte)
	2	Ratio Setpoint (high byte)
5	0	Exception Status
9	0	Ratio Readback (low byte)
	1	Ratio Readback
	2	Ratio Readback
	3	Ratio Readback (high byte)
10	0	Exception Status
	1	Ratio Switch
	2	Ratio Readback (low byte)
	3	Ratio Readback
	4	Ratio Readback
	5	Ratio Readback (high byte)
11	0	Exception Status
	1	Ratio Readback (low byte)
	2	Ratio Readback
	3	Ratio Readback
	4	Ratio Readback (high byte)
	5	Ratio Setpoint (low byte)
	6	Ratio Setpoint
	7	Ratio Setpoint
	8	Ratio Setpoint (high byte)
12	0	Ratio Switch
	1	Ratio Setpoint (low byte)
	2	Ratio Setpoint
	3	Ratio Setpoint
	4	Ratio Setpoint (high byte)
13	0	Sensor Value 1 (low byte)
	1	Sensor Value 1
	2	Sensor Value 1
	3	Sensor Value 1 (high byte)
	4	Sensor Value 2 (low byte)
	5	Sensor Value 2
	6	Sensor Value 2
	7	Sensor Value 2 (high byte)

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Table 36: Assembly Instance Details (continued)

Instance	Byte	Data
14	0	Exception Status
	1	Sensor Value 1 (low byte)
	2	Sensor Value 1
	3	Sensor Value 1
	4	Sensor Value 1 (high byte)
	5	Sensor Value 2 (low byte)
	6	Sensor Value 2
	7	Sensor Value 2
	8	Sensor Value 1 (low byte)
	9	Ratio Readback (low byte)
	10	Ratio Readback
	11	Ratio Readback
	12	Ratio Readback (high byte)

Connection Object: Class Code 05 (0x05)

The Connection Objects manage the characteristics of each communication connection. As a Group II Only Slave device the unit supports one explicit message connection and a POLL message connection.

Class Attributes

Table 37: Connection Object Class Attributes

Attribute	Access	Name	Type	Value
1	Get	Revision	UINT	1

Instance 1 Attributes (Explicit Message)

Table 38: Connection Object Instance 1 Attributes

Attribute	Access	Name	Type	Value
1	Get	State	USINT	See Below
2	Get	Instance Type	USINT	0 = Explicit Message
3	Get	Transport Class Trigger	USINT	0x83
4	Get	Production Connection	UINT	See Below
5	Get	Consumed Connection	UINT	See Below
6	Get	Initial Comm. Char.	USINT	0x21
7	Get	Production Size	UINT	128
8	Get	Consumed Size	UINT	128
9	Get/Set	Expected Packet Rate	UINT	default 2500 msec
12	Get/Set	Timeout Action	USINT	See Below
13	Get	Prod. Path Length	USINT	0
14	Get	Production Path		(null)
15	Get	Cons. Path Length	USINT	0
16	Get	Consumed Path		(null)

Instance 2 Attributes (Poll Connection)**Table 39: Connection Object Instance 2 Attributes**

Attribute	Access	Name	Type	Value
1	Get	State	USINT	See Below
2	Get	Instance Type	USINT	1 = I/O Message
3	Get	Transport Class Trigger	USINT	0x82
4	Get	Production Connection	UINT	See Below
5	Get	Consumed Connection	UINT	See Below
6	Get	Initial Comm. Char.	USINT	0x01
7	Get	Production Size	UINT	See Below
8	Get	Consumed Size	UINT	See Below
9	Get/Set	Expected Packet Rate	UINT	default 2500 msec
12	Get/Set	Timeout Action	USINT	See Below
13	Get	Prod. Path Length	USINT	6
14	Get/Set	Production Path	STRUCT of	
		Log. Seg., Class	USINT	0x20
		Class Number	USINT	0x04
		Log.Seg., Instance	USINT	0x24
		Instance Number	USINT	0x0A (default)
		Log.Seg., Attribute	USINT	0x30
		Attribute Number	USINT	0x03
15	Get	Cons. Path Length	USINT3	6
16	Get/Set	Consume Path	STRUCT of	
		Log. Seg., Class	USINT	0x20
		Class Number	USINT	0x04
		Log.Seg., Instance	USINT	0x24
		Instance Number	USINT	0x0C (default)
		Log.Seg., Attribute	USINT	0x30
		Attribute Number	USINT	0x03

Common Services**Table 40: Connection Object Common Services**

Service Code	Class	Instance	Service Name
05 (0x05)	Yes	Yes	Reset
14 (0x0E)	Yes	Yes	Get_Attribute_Single
16 (0x10)	No	Yes	Set_Attribute_Single

Connection States (Attribute 1)

- 0 = non-existent
- 1 = configuring
- 3 = established
- 4 = timed out

Produced and Consumed Connections (Attributes 4 and 5)

Connection 1 Produced Connection ID: 10xxxxxx011

Connection 1 Consumed Connection ID: 10xxxxxx100

Connection 2 Produced Connection ID: 01111xxxxxx

Connection 2 Consumed Connection ID: 10xxxxxx101

xxxxxx = Node Address.

Production and Consumed Sizes (Attributes 7 and 8)

The Production and Consumed sizes will change based on what Assembly Instance is chosen for the POLL request and POLL response. The Table below will define the Production and Consumed sizes will their respective Assembly.

Table 41: Connection Object Production and Consumed Sizes

Assembly	Production Size	Consumed Size
1	2	
2	4	
3	5	
4		3
5	1	
9	4	
10	6	
11	9	
12		5
13	8	
14	13	

Watch Dog Timeout Activity (Attribute 12)

- 0 = Timeout (I/O Messaging default)
- 1 = Auto Delete (Explicit Messaging, fixed value)
- 2 = Auto Reset

S-Device Supervisor Object: Class Code 48 (0x30)**Class Attributes****Table 42: S-Device Supervisor Object Class Attributes**

Attribute	Access	Name	Type	Value
1	Get	Revision	UINT	1

Instance 1 Attributes**Table 43: S-Device Supervisor Object Instance 1 Attributes**

Attribute	Access	NV	Name	Type	Value
3	Get	NV	Device Type	SHORT STRING	FRC
4	Get	NV	SEMI Standard Revision Level	SHORT STRING	E54-0997
5	Get	NV	Manufacturer's Name	SHORT STRING	MKS Instruments
6	Get	NV	Manufacturer's Model Number	SHORT STRING	Delta II
7	Get	NV	Software Revision Level	SHORT STRING	See Below
8	Get	NV	Hardware Revision Level	SHORT STRING	See Below
9	Get	NV	Manufacturer's Serial Number	SHORT_STRING	See Below
10	Get	NV	Device Configuration	SHORT_STRING	See Below
11	Get	V	Device Status	USINT	See Below
12***	Get	V	Exception Byte	BYTE	See Below
13	Get	V	Exception Detail Alarm	STRUCT of	
			Common Exception Detail	STRUCT of	
			Size	USINT	2
			Detail	ARRAY of	
13***	Get	V	Detail[0]	BYTE	See Below
13***	Get	V	Detail[1]	BYTE	See Below
			Device Exception Detail	STRUCT of	
			Size	USINT	1
			Detail	ARRAY of	

*** This attribute resides on the Controller's memory.

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Table 43: S-Device Supervisor Object Instance 1 Attributes (continued)

Attribute	Access	NV	Name	Type	Value
13***	Get	V	Detail[0]	BYTE	See Below
			Manufacturer Exception Detail	STRUCT of	
			Size	USINT	1
			Detail	ARRAY of	
13***	Get	V	Detail[0]	BYTE	See Below
14	Get	V	Exception Detail Warning	STRUCT of	
			Common Exception Detail	STRUCT of	
			Size	USINT	2
			Detail	ARRAY of	
14***	Get	V	Detail[0]	BYTE	See Below
14***	Get	V	Detail[1]	BYTE	See Below
			Device Exception Detail	STRUCT of	
			Size	USINT	1
			Detail	ARRAY of	
14***	Get	V	Detail[0]	BYTE	See Below
			Manufacturer Exception Detail	STRUCT of	
			Size	USINT	1
			Detail	ARRAY of	
14***	Get	V	Detail[0]	BYTE	See Below
15	Get/Set	NV	Alarm Enable	BOOL	0=Disable, 1=Enable
16	Get/Set	NV	Warning Enable	BOOL	0=Disable, 1=Enable
100	Get	V	Mode	USINT	0 = User, 2 = Factory
101	Get/Set	NV	Default Display	USINT	0 = Banner 1 = Ratio 2 = Sensor Value 4 = Temperature
103	Get	NV	IP Address	ARRAY of USINT[4]	255.255.255.255
*** This attribute resides on the Controller's memory.					

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Table 43: S-Device Supervisor Object Instance 1 Attributes (continued)

Attribute	Access	NV	Name	Type	Value
104	Get	NV	Subnet Mask	ARRAY of USINT[4]	255.255.255.255
105	Get	NV	Default Gateway	ARRAY of USINT[4]	255.255.255.255
106	Get	NV	Ethernet MAC Address	ARRAY of USINT[6]	FF:FF:FF:FF:FF:FF

Common Services

Table 44: S-Device Supervisor Object Common Services

Service Code	Class	Instance	Service Name
05 (0x05)	No	Yes	Reset
06 (0x06)	No	Yes	Start
07 (0x07)	No	Yes	Stop
14 (0x0E)	Yes	Yes	Get_Attribute_Single
16 (0x10)	No	Yes	Set_Attribute_Single

Object-Specific Services

Table 45: S-Device Supervisor Object-Specific Services

Service Code	Class	Instance	Service Name
75 (0x4B)	No	Yes	Abort
76 (0x4C)	No	Yes	Recover
78 (0x4E)	No	Yes	Perform_Diagnostics

Parameter	Form	Description
TestID	USINT	Type of diagnostic test to be performed

Test ID Parameter	Type of Diagnostics to be Performed
0	Standard

Manufacturer-Specific Services

These services do not exist, unless in Factory mode.

Table 46: S-Device Supervisor Object Common Services

Service Code	Class	Instance	Service Name
50 (0x32)	No	Yes	Change Mode

Parameter	Form	Description
Password	UINT	0x0000 = User mode ***** = Factory mode

Software Revision Level (Attribute 7)

The Software Revision will be a text string of the Major and Minor revision information of the Identity object. It will have the format XX.YYY, where XX is the major revision and YYY is the Minor revision. The revision code will match that provided by the Identity object.

Hardware Revision Level (Attribute 8)

The Hardware Revision will be a text string reflecting the current revision of the hardware. It will have the format XX.YYY, where XX is the major revision and YYY is the Minor revision.

Manufacturer's Serial Number (Attribute 9)

A string representation of the serial number in the Identity Object, Class 1, Instance 1, Attribute 6.

Device Configuration (Attribute 10)

A string representation of the configuration of the device. The default configuration will be "DEFAULT". If a value other than "DEFAULT" is used, then the device will not operate like the default unit.

Device Status (Attribute 11)

- 0 = Undefined
- 1 = Self Testing
- 2 = Idle
- 3 = Self-Test Exception
- 4 = Executing
- 5 = Abort
- 6 = Critical Fault

Exception Status (Attribute 12)**Table 47: S-Device Supervisor Object Exception Status**

Bit	Function
0	ALARM/ device-common
1	ALARM/ device-specific
2	ALARM/ manufacturer-specific
3	Reserved – set to 0
4	WARNING/ device-common
5	WARNING/ device-specific
6	WARNING/ manufacturer-specific
7	1 = Expanded Method

Common Exception Detail (Attributes 13 and 14)**Table 48: S-Device Supervisor Object Common Exception Detail**

Bit	Function	Bit	Function
0	Internal diagnostic exception	0	0
1	Microprocessor exception	1	0
2	EPROM exception	2	0
3	EEPROM exception	3	Power supply input voltage
4	RAM exception	4	0
5	0 - Reserved by Devicenet	5	Notify manufacturer
6	Internal real-time exception	6	Reset exception
7	0 = Reserved by Devicenet	7	0 = Reserved by Devicenet

Device Exception Detail (Attributes 13 and 14)**Table 49: S-Device Supervisor Object Exception Status**

Bit	FRC Device Exception Detail
0	Flow Ratio Error S-Single Stage Controller
1	Reserved = 0
2	Reserved = 0
3	Reserved = 0
4	Reserved = 0
5	Reserved = 0
6	Reserved = 0
7	Reserved = 0

Manufacturer Exception Detail (Attributes 13 and 14)**Table 50: S-Device Supervisor Object Manufacturer Exception Detail**

Bit	Manufacturer Exception Detail
0	Reserved = 0
1	Reserved = 0
2	Flow 2 Overrange S-Analog Sensor (2)
3	Flow 2 Underrange S-Analog Sensor (2)
4	Flow 1 Overrange S-Analog Sensor (1)
5	Flow 1 Underrange S-Analog Sensor (1)
6	0 = Reserved
7	Internal Communication Fault

S-Analog Sensor Object Class Code: 49 (0x31)**Class Attributes****Table 51: S-Analog Sensor Object Class Attributes**

Attribute	Access	Name	Type	Value
1	Get	Revision	UINT	1
2	Get	Max Object Instance	UNIT	2

Instance 1-2 Attributes**Table 52: S-Analog Sensor Object Instance 1-2 Attributes**

Attribute	Access	NV	Name	Type	Value
3	Get/Set	NV	Data Type	USINT	0xC3=INT, 0xCA=REAL
4	Get/Set	NV	Data Units	UINT	0x1001 = counts, 0x1007 = %Full Scale, 0x1400 = sccm, 0x1401 = slm
5***	Get	V	Reading Valid	BOOL	0=invalid, 1=valid
6***	Get	V	Value (Flow)	(See Data Type)	See Below
7***	Get	V	Status	BYTE	See Below
10***	Get	NV	Full Scale	(See Data Type)	See Below
103***	Get	V	Q Zero State	USINT	See Below
*** This attribute resides on the Controller's memory.					

Common Services

Table 53: S-Analog Sensor Object Common Services

Service Code	Class	Instance	Service Name
14 (0x0E)	Yes	Yes	Get Attribute Single
16 (0x10)	No	Yes	Set_Attribute_Single

Object-Specific Services

Table 54: S-Analog Sensor Object-Specific Services

Service Code	Class	Instance	Service Name
None: There are no Object-Specific Services.			

Vendor-Specific Services

Table 55: S-Analog Sensor Object Vendor-Specific Services

Service Code	Class	Instance	Service Name
50 (0x32)	No	Yes	Zero Unit

This service takes no data. (i.e. just send service code, class code, and instance code).

This is used to perform a zero offset adjust service on the Delta II. Sending down this service code, calculates a zero offset for BOTH sensors. Use either instance 1 or 2 to perform this service.

This will take several seconds to perform. The device should be in the “Normal control mode”(Single Stage Controller Object, attribute 5,) and executing state. There should be < 82 counts(2% FS) of flow in either sensor or the adjustment will not be performed.

The offsets are stored in the Offset-B attribute. Monitoring the “Zero State” attribute will show the progress.

Value (Flow) (Attribute 6)

The corrected, converted, calibrated final value of the sensor.

Status (Attribute 7)

Table 56: S-Analog Sensor Object Status

Bit	Function
0	High Alarm Exception
1	Low Alarm Exception
2	High Warning Exception
3	Low Warning Exception
4	0 – Reserved
5	0 – Reserved

6	0 – Reserved
7	0 – Reserved

Full Scale (Attribute 10)

The value of attribute Value corresponding to the Full Scale calibrated measurement of the sensor. Default value will be maximum allowable value for the Data Type.

Q Zero State (Attribute 103)

Instance 1 is for Q1 Zero State, Instance 2 is for Q2 Zero State.

Table 57: S-Analog Sensor Object Q Zero State

Value	State
0	Complete Success
1	Complete Unsuccessful
2	In Progress

S-Single Stage Controller Object: Class Code 51 (0x33)**Class Attributes****Table 58: S-Single Stage Controller Object Class Attributes**

Attribute	Access	Name	Type	Value
1	Get	Revision	UINT	1

Instance 1 Attributes**Table 59: S-Single Stage Controller Object Instance 1 Attributes**

Attribute	Access	NV	Name	Type	Value
3	Get/Set	NV	Data Type	USINT	0xC3=INT, 0xCA=REAL
5	Get/Set	NV	Control Mode	USINT	See Below
6	Get/Set	V	Setpoint	(See Data Type)	See Below
7***	Get/Set	V	Process Variable	(See Data Type)	See Below
10***	Get	V	Status	BYTE	See Below
11	Get/Set	NV	Alarm Enable	BOOL	0 = disabled, 1 = enabled
12	Get/Set	NV	Warning Enable	BOOL	0 = disabled, 1 = enabled
13	Get/Set	NV	Alarm Settling Time	UINT	See Below
14	Get/Set	NV	Alarm Error Band	(See Data Type)	See Below
15	Get/Set	NV	Warning Settling Time	UINT	See Below
16	Get/Set	NV	Warning Error Band	(See Data Type)	See Below
17	Get/Set	NV	Safe State	USINT	See Below
18	Get/Set	NV	Safe Value	(See Data Type)	See Below

107	Get/Set	NV	Default Ratio Switch	USINT	See Below
108	Get/Set	V	Ratio Switch	USINT	See Below
*** This attribute resides on the Controller's memory.					

Common Services

Table 60: S-Single Stage Controller Object Common Services

Service Code	Class	Instance	Service Name
14 (0x0E)	Yes	Yes	Get_Attribute_Single
16 (0x10)	No	Yes	Set_Attribute_Single

Control Mode (Attribute 5)

Table 61: S-Single Stage Controller Object Control Mode

Attribute Value	State
0	Normal (Normal closed loop operation)
1	Zero/Off/Closed (Both valves closed)
2	Full/On/Open (Both valves open)
3	Hold (Do not change any valve position, hold current positions)
4	Safe State (Go to state defined 'safe')
5-63	Reserved
64-127	0
132	Open Close Mode (Q ₂ valve open, Q ₁ valve closed if Ratio Switch set for Q ₂ /Q ₁ operation, otherwise vice versa.)
128-255	0

Setpoint (Attribute 6)

The Setpoint sent to the device actuator. This is the desired ratio in which you want the process variable driven to.

Process Variable (Attribute 7)

The measured ratio output.

Status (Attribute 10)

Table 62: S-Single Stage Controller Object Status

Bit	Definition
0	Alarm Exception
1	Warning Exception
2	0 – Reserved
3	0 – Reserved
4	0 – Reserved

5	0 – Reserved
6	0 – Reserved
7	0 – Reserved

Alarm Settling Time (Attribute 13)

The number of milliseconds allowed for the control-loop to settle to within the error band.

Alarm Error Band (Attribute 14)

The amount by which the Setpoint must equal the Process Variable, or an alarm condition will occur.

Warning Settling Time (Attribute 15)

The number of milliseconds allowed for the control-loop to settle to within the error band.

Warning Error Band (Attribute 16)

The amount by which the Setpoint must equal the Process Variable, or a warning condition will occur.

Safe State (Attribute 17)

This attribute is what will be used to control the device when the device is in a state other than executing.

Table 63: S-Single Stage Controller Object Status

Attribute Value	State
0	Zero/Off (Both valves closed)
1	Full Scale/On (Both valves open)
2	Hold Last Value
3	Use Safe Value
4-63	Reserved
64-127	0
128-255	0

Safe Value (Attribute 18)

For Safe State set to Use Safe Value, the ratio will be driven to this value.

Valve Select (Attribute 100)

This attribute does not exist, unless in Factory mode.

Valve Select for fine control Q1 or Q2 side.

Fixed Valve Q2 High (Attribute 101)

This attribute does not exist, unless in Factory mode.

Valve control offset. For “fixed” valve.

Kd, Ki, and Kp (Attributes 102-104)

This attribute does not exist, unless in Factory mode.

The PID loop constants for both valves.

Default Ratio Switch (Attribute 107)

This is the value that the Ratio Switch attribute will default to on power-on and resets.

Default is Q2/Q1, selectable to Q1/Q2.

Ratio Switch (Attribute 108)

Determines the flow stream channel that is controlling. Ratio will either be Q2/Q1 or Q1/Q2.

Value of 1 = Q1/Q2

Value of 2 = Q2/Q1

Default Attribute Values**Table 64: Default Attribute Values**

Class	Instance	Attribute	Value
48	1	8	“01.001”
48	1	10	Default
48	1	15	1 = Enabled
48	1	16	1 = Enabled
49	1 and 2	3	0xCA = REAL
49	1 and 2	4	0x1400 = sccm
51	1	3	0xCA = REAL
51	1	5	0 = Normal
51	1	11	0 = disabled
51	1	12	0 = disabled
51	1	13	10000
51	1	14	1.0%
51	1	15	5000
51	1	16	1.0%
51	1	17	1 = On/Open
51	1	18	0.00

How to Override the Valves

The valve override feature enables the proportional control valves to be fully opened (purged) or closed independent of the ratio set point command signal. Refer to *Devicenet Protocol*, page 55, for additional information.

To *open* the valves, send the following Devicenet command:

Table 65: Devicenet Command to Open the Valves

Object	Instance	Attribute	Service	Value
0x33	1	5	0x10	2

To *close* the valves, send the following Devicenet command:

Table 66: Devicenet Command to Close the Valves

Object	Instance	Attribute	Service	Value
0x33	1	5	0x10	1

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Chapter Seven: Maintenance and Troubleshooting

General Information

In general, no maintenance is required other than proper installation and operation, zero adjustment, periodic calibration, checks for wear on the cables and inspection of the enclosure for visible signs of damage. If an MKS Delta II Ratio[®] Controller fails to operate properly upon receipt, check for shipping damage, and check the power/signal cable for correct continuity, grounding and voltage levels. Any damage should be reported to the carrier and MKS Instruments immediately. If there is no obvious damage and the continuity is correct, obtain an RMA Number (Return Material Authorization Number) and complete a Health and Safety Form before returning the unit to MKS Instruments for service.

How to Zero the MKS DELTA II Ratio Controller

For best accuracy and repeatability, the zero setting periodically needs to be set. The frequency of resetting the zero is dependent on the specific accuracy and repeatability required by your process. It is also recommended that the instrument be recalibrated annually if no other time interval has been specifically established. Refer to the inside back cover of this instruction manual for a complete list of MKS Calibration and Service centers.



Warning

If the instrument is being used to control dangerous gases, be sure that the system is *fully warmed up* before applying gases to the system. You may choose to install a positive shutoff valve to prevent inadvertent gas flow during the warm-up period.

Once the MKS Delta II Flow Ratio Controller is completely warmed up, you can proceed to zero the unit as required.

Overview

The Zero adjust feature may be implemented either via the Devicenet connection or via the embedded HTML web based UI. Implementation via Devicenet is covered in this section. For implementation using the embedded UI, see Chapter Four, page 35.

The electrical *Zero adjust* feature integrated into the MKS Delta II Flow Ratio[®] Controller is a digital equivalent to the *zero adjust* potentiometer or “autozero” pins found on many analog mass flow controllers. The feature allows the user to compensate for long term drift and attitude sensitivity, universally present in thermal sensor based mass flow instruments. 1% of full scale zero drift and 1% of reading drift per year are typical flow channel specifications. Similarly, attitude sensitivity can result in up to 1% of reading “offsets” due to thermal siphoning effects.

The components used in the MKS Delta II Flow Ratio[®] Controller have a long history of usage and are expected to comply with the specifications given above. Although the Delta II does not allow for direct flow measurements and uses a differential technique which tends to 'automatically' cancel drifts in the sensors that are the same in both channels, there are times when it is beneficial to do so. See *When to Use*, below.

When to Use

Use of the *zero adjust* feature is appropriate when:

- The installation orientation is changed from the calibrated (horizontal, base down) orientation.
- When ambient temperatures at installation are $>\pm 10^\circ$ from the temperature at calibration (Nom. 20°C).
- It is included as part of a semi-annual or annual preventive maintenance activity.
- If process “drift” is identified that cannot be attributed to other causes, applying *zero adjust* to the Flow Ratio Controller can be used as part of diagnostic efforts.

It is not recommended to use *zero adjust* each time the instrument is idle; every 6 months to a year between zero adjusts should be more than adequate if the device has been left under continuous power. Flow Ratio Controllers are currently calibrated in a horizontal, base down configuration and should not require any *zero adjust* on installation in this orientation.

How to Use

Ensure that no gas flow is entering the ratio flow controller during the zero operation.

To implement zero adjustment, the MKS Delta II Flow Ratio® Controller must be isolated from the system, typically done by closing a positive shutoff valve downstream of the instrument. If an upstream valve is available, it should be closed as well. The unit should be above 100 Torr when zeroed.

1. Give a valve open command to the Delta II.
2. Apply gas, at a regulated pressure, to the ratio flow controller.
3. Close the downstream positive shutoff valve.
4. Flow gas to raise pressure in the Delta II above 100T.
5. Close the upstream positive shutoff valve and let unit sit for 1-2 minutes to equalize.
6. Zero the unit.

Zero the Unit

1. Have the unit perform an automated zero adjust by sending a Zero Adjust service to the device.
2. Wait 20 seconds while the controller performs the zero read and zero adjustment storage.
3. Open the positive shut off valve(s) in the system.

Both channels of the Flow Ratio Controller are now re-zeroed and the unit is restored to control mode, ready for use. Cycling power is not necessary for the new values to be in effect.

Note that the MKS Delta II Flow Ratio® Controller may indicate a ratio even a very large one ratio even when the Disable Switch command (which closes proportionate valves in the instrument) is used due to closed conductance leakage across its proportional control valves. A small change in flow (even if due to signal noise) across one channel can have a significant effect on the reported ratio in these conditions.

However, do not “zero out” this ratio flow since it represents an actual ratio flow measurement inherent in the system.

Notes:

- The *zero adjustment* is executed by the instrument by comparing current sensor readings with the zero flow reading stored during calibration. The difference in these values is stored in a unique register in the instrument EEPROM. On start-up, this value is added or subtracted to all values in the calibration table and the resultant tables are loaded into RAM for unit operation.
- Because this operation offsets the entire calibration table for both sensors, **the user must be certain that a zero flow condition exists** before enabling the function.

- **If the difference between the “at calibration” and “at zero adjust” sensor values for either channel is greater than 3% of channel full scale, zero adjust will not be performed.**
- Since the Flow Ratio Controller is not equipped with independent flow output signals, this is a “blind” operation. The feature has undergone rigorous testing and behaves per the design intent (same as an automatic zero feature on Delta II Ratio Controller’s).

Repair

Contact any authorized MKS Sales Office or Calibration and Service Center should you encounter any difficulties or problems using your MKS Delta II Flow Ratio Controller.



Note

If it is necessary to return the instrument to MKS for repair, please contact any of the MKS international service/calibration centers listed on the inside of the back cover of this manual for an RMA (Return Material Authorization) number to expedite handling and ensure proper servicing of your instrument.

Troubleshooting**Table 67: Troubleshooting Chart**

Symptoms	Possible Cause	Remedy
No ratio output or overrange (after warm-up)	Improper cable.	Check cable for type/grounding per instructions.
	Device has not received “Start” command and/or control mode is not set to Normal.	Send “Start” command and set control mode to “Normal” mode.
	One of the valve override functions applied (Disable Switch or Valve Open).	Disconnect valve overrides.
	No gas flow.	Provide gas.
	Electronics malfunctioning.	Return for service.
Controller does not track set point	Device has not received “Start” command and/or control mode is not set to Normal.	Send “Start” command and set control mode to “Normal” mode.
	Operation out of flow design range for either channel.	Ensure flow through both channels are within design parameters.
	Downstream conductance differences.	Check downstream plumbing for similarities and differences relative to flow conductance.
	Improper zero adjustment.	Zero Delta II.
Controller does not function	Electronics malfunctioning.	Return for service.
Oscillation	Incorrect upstream pressure regulator.	Check manufacturers’ specifications.
	Upstream pressure too high.	Reduce upstream pressure.
	Downstream conductance differences.	Check downstream plumbing for similarities and differences relative to flow conductance.
	Another active control device downstream of one of the output channels.	Remove the other active device.
Unit does not achieve full flow	Upstream pressure too low.	Increase upstream pressure.

Appendix A: Product Specifications

Performance Specifications

Ratio Accuracy ¹	± 2% Setpoint
Ratio Flow Control Range per Channel	5 to 100% F.S.
Ratio Repeatability	±0.3% of Setpoint
Resolution	0.02% of channel FS
Ratio Range	1:1 to 20:1 and 20:1 to 1:1 (using Ratio switch)
Maximum Operating Outlet Pressure	200 Torr
Maximum Allowable Outlet Pressure Differential	50 Torr
Normal Operating Pressure Differential	<150 Torr
CE Compliance ² Electromagnetic Compatibility	EMC Directive 89/336/EEC
Ratio Settling Time	<2 seconds
Full Scale Ranges (nitrogen equivalent – Q2/Q1)	500/500, 1K/1K, 2K/2K, 10K/10Kscm
Maximum Inlet Pressure	150 psig (non-operational)
Temperature Coefficients Zero Span	< 0.05% F.S./° C (500 ppm) <0.08% of Rdg/° C (800 ppm)
Warm up Time	60 minutes
Normal Operating Temperature	10° to 40° C
Storage Temperature	-20° to 65° C
Temperature Display	0° to 100° C
Temperature Readout Units	°C
Temperature Accuracy	±2° C
Temperature Resolution	0.1° C
¹ Includes non-linearity, hysteresis, and non-repeatability.	
² An overall metal braided, shielded cable, properly grounded at both ends, is required during use.	

Environmental Specifications

Storage Humidity Range	0 to 95% relative humidity, non-condensing
Storage Temperature	-20° to 65° C (-4° to 122° F)

Electrical Specifications

Connector/Control Options	
Digital I/O	5-pin DeviceNet Micro Connector Type
Analog I/O	9-pin D male
	RJ-45 female
Input Voltage	11 to 25 VDC (Devicenet compatible)
Input Voltage/Current Required	
Maximum at Start Up (first 5 seconds)	+24 VDC ($\pm 5\%$) @ 1 A
Typical at Steady State	+24 VDC ($\pm 5\%$) @ 375 mA

Physical Specifications

Body (height x width x length) <i>overall</i>	5 in x 3.2 in x 4.88 in 12.70 cm x 8.13 cm x 17.53 cm
Fittings	
Inlet	Swagelok compatible 4-VCR male type, non-rotatable
Outlets x 2	Swagelok compatible 4-VCR male type, non-rotatable
Leak Integrity	
External (scc/sec He)	$< 1 \times 10^{-9}$
Through closed valve (Closed Conductance)	$< 2.0\%$ F.S. @ 2 psig inlet to atm outlet
Materials Wetted	
Body and seals	316L SST, Inconel, Elgiloy
Valve seat	Teflon
Surface Finish	$< 5 \mu$ inches Ra, average – Electropolished
Weight	4.65 lbs (2.1 kg)

Due to continuing research and development activities, these product specifications are subject to change without notice.

Appendix B: Model Code Explanation

Model Code

The options of your MKS Delta II Flow Ratio Controller unit are identified in the model code when you order the unit.

The model code is identified as follows: **DLT2A2 XX Y B C A Z QQ**
where:

	DLT2A2	XX	Y	B	C	A	Z	QQ
Type Number								
Full Scale Range								
Full Scale Ratio								
Interface Electronics								
Signal I/O Electronics								
Control Type								
Control Channel								
Firmware Version								

Type Number (DLT2A2)

This designates the model number of the MKS Delta II Flow Ratio Controller instrument.

Full Scale Range: sccm of Nitrogen for Q2 channel (XX)

The full scale range for Q2 is indicated by a two digit code.

Full Scale Range	Ordering Code
500	52
1000	13
2000	23
10000	14

Full Scale Ratio Q2/Q1 Ration (Y)

This code indicates the ratio of FS flow available in the unit. All units are available in a 1:1 full scale configuration - the full scale flow for both channels are the same. For example, a 500/500 unit is a 1:1 FS unit.

Full Scale Ratio	Ordering Code
1:1	1

* Consult the factory for availability of other configurations.

Interface Electronics (B)

The MKS Delta II Flow Ratio Controller is available in two interface configurations: Analog and Devicenet (also called Digital). This code specifies the type.

Interface Electronics	Ordering Code
Analog Interface Electronics - 9 pin D connector	A
DeviceNet Interface Electronics – 5 pin microconnector	6

Signal I/O Electronics (C)

There is a choice of signal I/O options available with the Analog version MKS Delta II Flow Ratio Controller. This specifies the range of operation that the unit will accept for ratio control and output for ratio readback. Note that the Devicenet version does not specify any Signal I/O options.

Signal I/O Voltage	Ordering Code
Analog 0V – 5V	1
Analog 0V – 10V	2
DeviceNet	3

* The user of analog devices may reconfigure this using the Configuration page of the embedded UI of the Delta II.

Control Type (A)

There is a choice of controlling based on ratio or on percentage flow relative to the control channel.

Control Type	Ordering Code
Ratio: Q_2/Q_1 or Q_1/Q_2	R*
Percentage: $Q_2/(Q_1+Q_2)$ or $Q_1/(Q_1+Q_2)$	P**
DeviceNet Units must select R	

** The users of analog devices may reconfigure this using the Configuration page of the embedded UI of the Delta II.

Control Channel (Z)

There is a choice of controlling based on ratio or on percentage flow relative to the control channel.

Control Channel	Ordering Code
Q1 Control: Q_1/Q_2 or $Q_1/(Q_1+Q_2)$	1
Q2 Control: Q_2/Q_1 or $Q_2/(Q_1+Q_2)$	2*

* The user may reconfigure this using the Configuration page of the embedded UI of the Delta II.

Firmware Version (QQ)

The firmware version on the device is a two digit numerical code. Prior to product release this is coded AA.

Firmware Version	Ordering Code
Firmware Version	AA

How to Order an MKS DELTA II Flow Ratio Controller

To order an analog 500/500 sccm MKS Delta II Flow Ratio Controller with Devicenet the product code is:

DLT2A252163R2AA

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Appendix C: DELTA II Sizing Guidelines

General Information

To select the correct MKS Delta II Flow Ratio Controller for an application, you must:

- Identify all gas types and all flow rates and conditions required. This requires all combinations of gas and flow rates along with expected ratio/percentage control range is desired.
- Use MKS Delta II Range Calculator[®] to select proper Delta II model.
- Determine what types of interface I/O are required.

How to Determine the Flow Controller Range

The MKS Delta II Flow Ratio Controller is available with full-scale channel flow rates in the ranges of 500, 1000, 2000, and 10,000 sccm (N₂ equivalent). To select the appropriate range, you must determine the flow rate of nitrogen that is equivalent to the flow rate of the desired gas or gas mixture. Calculate the ratio of the GCF of nitrogen (1.00) to the GCF of the desired gas as shown in the following example.

Example:

You need a total flow rate of 750 sccm of argon (Ar) and a ratio of (9:1) to (1:1). What range flow ratio controller should you use?

1. Find the Gas Correction Factor of argon. The GCF for Ar is 1.4.
2. Insert the GCF of Ar in the following formula:

$$\frac{(\text{GCF of N}_2)}{(\text{GCF of Ar})} = \frac{(x)}{(\text{Desired flow rate of Ar})}$$

where x is the equivalent flow rate of nitrogen (sccm).

$$\frac{(1.00)}{(1.4)} = \frac{(x)}{(750 \text{ sccm Ar})}$$

$$x = 536 \text{ sccm N}_2$$

A flow rate of 750 sccm of Ar will produce a flow rate equivalent to 536 sccm of N₂. This falls within the range of a 500 sccm channel ratio flow controller with 268 sccm through each channel for 1:1 and 485/54 sccm for 9:1.

When calculating equivalent N₂ flow using gas correction factors, be sure to use a ratio controller with a sufficient flow rate range. For example, if the calculated equivalent N₂ flow in the example shown above is 1200 sccm, use a 2000 sccm channel ratio flow controller.



Note

When using a gas with a density *higher* than nitrogen, be sure that the control valve Full Scale range can accommodate the desired flow rate. Please call the MKS Applications group if you have any questions.

The ratio of control falls within the specified operation range of a 500/500 unit.

Now add the requirement that the unit must also control of 1000sccm (N₂ Equivalent) of a gas (or mixture of gases) from (9:1) to (3:1). The first condition above is considered the 'low flow' condition. This added requirement is the 'high flow' condition. Since the 'high flow' requirement falls outside the limits of a 500/500 unit (at 1:1 and above, the flow in one channel is above rated FS of unit), at least a 1000 sccm unit must be used. A 1000/1000 unit can be used because some of the ratio conditions from the 'low flow' condition also fall within the operating limits. For example, at a ratio of 9:1 in the 'low flow' condition, only about 54 sccm (10% of 536) will be flowing through a channel. This is only 5.4% of flow for the 1000 sccm channel.

Given the above, it is easily seen that determining the operating regime for a multi-gas, multi-flow and multi-ratio set of conditions may be somewhat complex. To simplify this task, MKS has developed a Delta II Range calculator. To make the most use of this calculator, it is recommended that the potential user create a table such as that below for the conditions under which the controller is expected to operate. For each process step, the calculator can then be used to determine whether the conditions are within the range of the device.

Table 68: Process Gas Recipe Table

GAS	GAS FLOW RATE (sccm of the gas)					
	Step 1	Step 2	Step 3	Step 4	Step n-1	Step n
Gas 1						
Gas 2						
Gas 3						
Gas 4						
Gas n-1						
Gas n						
Ratio/Percentage						
Delta II Range Calculator will determine the data in the section below based on above inputs						
Total Gas Flow (Nitrogen equivalent)						
Flow Channel 1						
Flow Channel 2						

Using the MKS Delta II Range Calculator

MKS has available for use a spreadsheet that can be used to help in the selection of the correct ratio flow controller to fit your specific application. Contact the factory for more information.

Appendix D: Health and Safety Form



HEALTH AND SAFETY FORM

THIS FORM MUST BE COMPLETED AND RETURNED WITH EQUIPMENT OR SERVICE WILL NOT BE PERFORMED

RETURN MATERIAL AUTHORIZATION NUMBER (RMA#):	
RETURN TO STOCK NUMBER/RTS# (If applicable):	Trade in number (if applicable):

Section 1: (one instrument per form)	MKS Part Number:
	MKS Serial Number:

Section 2: Has this equipment been used? *(Please check appropriate boxes)*

☐	No – Still in MKS packaging
☐	No – Unit unpacked, but never installed in a system.
☐	Yes -- Used only with clean, dry inert gas (For Example: Air, N2, Ar, He).
☐	Yes -- Used with chemicals, non-inert gases, biological or radioactive agents.) Identify all materials:
☐	Yes -- Used in a Semiconductor Copper process. Equipment must be double bagged. Label outside bag and packing slip, Copper Part. Label final shipping container Copper Part and place a strip of ORANGE TAPE on the container.
	Has equipment been purged? ☐ No ☐ yes purged with what?
	Has equipment been flushed? ☐ No ☐ yes flushed with what?
	Has equipment been decontaminated? ☐ no ☐ yes, explain process:
	How many months in use?

Section 3: Detailed failure information or description or required service or reason for return.

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Section 4: Company or Organization (mandatory information)

Company:		
Address:		
City:	State:	Zip:
Printed Name:	Signature:	
Date:	Phone #:	
Email:	Fax #:	
End User (if applicable):		

For MKS USE only:

MKS Subsidiary or Agent:
Contact Name:
Customer #:
Maximum Credit allowed (TBD after inspection)

ALL PRODUCTS MUST BE RETURNED IN SEALED BAGS

MKS will not accept delivery of equipment that has been chemically, radioactively or biologically contaminated, without written evidence of decontamination or laboratory analysis. Alternately, we will require evidence that the biological process is not harmful.

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